Probability and Stochastic Processes

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Chapter 3. Random Vectors

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Random Vectors

We are often interested in probability statements concerning two or more random variables.

For example, if the experiment is your physical examination, you may find out your height H and weight W. These are separate random variables, but are often informative when considered jointly.

That means we should think of H and W as being the components of a "random vector" (H, W) taking values in $\mathbb{R}^2$, rather than being unrelated random variables each taking values in $\mathbb{R}$.

Likewise, the random vector comprising height, weight, age, sex, blood pressure, and heart rate is of more use considered jointly than separately.

Random Vectors

There are two efficient tools to study random vectors.

One is studying the joint distribution, which gives the probability of each possible pair of values for the random vector.

Another is observing numerical characteristics, which can give us information of the relationship between random variables. As you can imagine, to handle the problems of random vectors must be getting more complicated.

Section 3.1 The Distribution Function of a Random Variable

- 3.1.1 Random Vectors and Joint Distributions
- 3.1.2 Discrete Random Vectors
- 3.1.3 Continuous Random Vectors

Random Vectors and Joint Distributions

Let X and Y be two random variables defined on the same probability space. The distribution functions $F_X(x)$ and $F_Y(y)$ of the given random variables only determine their separate (marginal) statistics. How can we describe their joint statistics? In this section, we show that the joint statistics of the random variables X and Y can be completely determined by their joint distribution function.

Random Vectors and Joint Distributions

The ***joint distribution function (joint d.f.)*** of X and Y is defined by

$$F(x, y) = P(X \leqslant x, Y \leqslant y), \quad -\infty < x, y < \infty. \tag{1}$$

All joint probability statements about X and Y can, in theory, be answered in terms of their joint distribution function by using the equation (1). For example,

Random Vectors and Joint Distributions

(i) for any $x_1 < x_2, y_1 < y_2$, the event $\{X \leqslant x, y_1 < Y \leqslant y_2\}$, we maintain that

$$P(X \leqslant x,\ y_1 < Y \leqslant y_2) = F(x, y_2) - F(x, y_1);$$

(ii) for the event $\{x_1 < X \leqslant x_2, Y \leqslant y\}$, we maintain that

$$P(x_1 < X \leqslant x_2, Y \leqslant y) = F(x_2, y) - F(x_1, y);$$

(iii) for the event $\{x_1 < X \leqslant x_2, y_1 < Y \leqslant y_2\}$, we have

$$\begin{aligned} &P(x_1 < X \leqslant x_2, y_1 < Y \leqslant y_2) \\ =&F(x_2, y_2) - F(x_1, y_2) - F(x_2, y_1) + F(x_1, y_1). \end{aligned}$$

Random Vectors and Joint Distributions

Property

The joint distribution function $F(x, y)$ *is such that*
(i) *for any* $x_1 < x_2$, $F(x_1, y) \leqslant F(x_2, y)$; *for any* $y_1 < y_2$, $F(x, y_1) \leqslant F(x, y_2)$.
(ii) $F(-\infty, y) = 0, \quad F(x, -\infty) = 0, \quad F(\infty, \infty) = 1.$
(iii) $\lim\limits_{x \to x_0^+} F(x, y) = F(x_0, y), \qquad \lim\limits_{y \to y_0^+} F(x, y) = F(x, y_0).$

A natural question is what values $F(x, +\infty)$ and $F(+\infty, y)$ are.

First, the values can not be 1. In fact, they are just the distributions of single random variables X and Y respectively.

Random Vectors and Joint Distributions

The d.f. of the single variable X can be derived from the joint d.f. $F(x, y)$ as follows, for $-\infty < x < \infty$,

$$\begin{aligned} F_X(x) &= P(X \leqslant x) = P(X \leqslant x, Y < \infty) = F(x, \infty) \\ &= P(\lim_{y \to \infty} \{X \leqslant x, Y \leqslant y\}) \\ &= \lim_{y \to \infty} P(X \leqslant x, Y \leqslant y) = \lim_{y \to \infty} F(x, y). \end{aligned}$$

$$i.e., \quad F_X(x) = \lim_{y \to \infty} F(x, y).$$

Similarly, if $F_Y(y)$ denotes the $d.f.$ of Y, then for $-\infty < y < \infty$,

$$F_Y(y) = \lim_{x \to \infty} F(x, y).$$

The $d.f.$ F_X and F_Y are referred to as the ***marginal distributions*** of X and Y.

Random Vectors and Joint Distributions

For instance, we wanted to compute the joint probability that X is greater than x and Y is greater than y. This could be done as follows.

$$
\begin{aligned}
& P(X > x, Y > y) \\
&= 1 - P(\overline{\{X > x, Y > y\}}) \\
&= 1 - p(\overline{\{X > x\}} \cup \overline{\{Y > y\}}) \\
&= 1 - P(\{X \leqslant x\} \cup \{Y \leqslant y\}) \\
&= 1 - [P(X \leqslant x) + P(Y \leqslant y) - P(X \leqslant x, Y \leqslant y)] \\
&= 1 - F_X(x) - F_Y(y) + F(x, y).
\end{aligned}
$$

Random Vectors and Joint Distributions

Example

The joint distribution function of X and Y is

$$F(x,y) = A\Big(B + \arctan\frac{x}{2}\Big)\Big(C + \arctan\frac{y}{2}\Big), \quad -\infty < x, y < \infty.$$

(a) Determine the values of the constants A, B and C.
(b) Determine the value of $P(0 < X \leqslant 2,\ 0 < Y \leqslant 3)$.
(c) Find the marginal distributions of X and Y.

Solution. (a) Since $F(\infty, \infty) = 1$,

$$F(\infty, \infty) = A(B + \pi/2)(C + \pi/2) = 1.$$

Since $F(-\infty, y) = 0$ and $F(x, -\infty) = 0$,

$$F(-\infty, y) = A(B - \pi/2)\Big(C + \arctan\frac{y}{2}\Big) = 0$$

and

$$F(x, -\infty) = A\Big(B + \arctan\frac{x}{2}\Big)(C - \pi/2) = 0.$$

Thus we have $B = C = \dfrac{\pi}{2}$ and $A = \dfrac{1}{\pi^2}$.

(b) From equation (3.2), we know

$$P(0 < X \leqslant 2,\ 0 < Y \leqslant 3)$$

$$= F(0,0) + F(2,3) - F(0,3) - F(2,0) = 1/16.$$

Solution. (c) The marginal distributions of X and Y are

$$F_X(x) = \lim_{y\to\infty} F(x,y) = \frac{1}{\pi}\Big(\frac{\pi}{2} + \arctan\frac{x}{2}\Big),$$

and

$$F_Y(y) = \lim_{x\to\infty} F(x,y) = \frac{1}{\pi}\Big(\frac{\pi}{2} + \arctan\frac{y}{2}\Big)$$

respectively. $\square$

Section 3.1 The Distribution Function of a Random Variable

- 3.1.1 Random Vectors and Joint Distributions
- 3.1.2 Discrete Random Vectors
- 3.1.3 Continuous Random Vectors

Discrete Random Vectors

For discrete bivariate random vectors, we always use the ***joint probability function*** (joint $p.f.$)

$$p(x, y) := P(\{X = x\} \cap \{Y = y\}) \equiv P(X = x, Y = y)$$

of the pair of discrete random vector (X, Y), whose possible values are a (finite or countably infinite) set of points (x_j, y_k) in the plane, where $p(x, y)$ has the following properties:

(i) $p(x_j, y_k) \geqslant 0$, $\forall (x_j,\ y_k) \in \mathbb{R} \times \mathbb{R}$;

(ii) $\displaystyle\sum_{j=1}^{\infty} \sum_{k=1}^{\infty} p(x_j, y_k) = 1.$

Discrete Random Vectors

From the definition of $p(x, y)$, we know

$$F(x, y) = \sum_{x_j \leqslant x} \sum_{y_k \leqslant y} p(x_j, y_k). \tag{2}$$

The probability $P[(X, Y) \in A]$ is obtained by summing the joint p.f. over pairs in A:

$$P[(X, Y) \in A] = \sum_{(x_j, y_k) \in A} p(x_j, y_k).$$

When the function $p(x, y)$ is summed over all possible values of Y (resp., X), the resulting function is called the ***marginal probability function*** of X (resp., Y). That is,

$$p_X(x) = \sum_y p(x, y) \quad \text{and} \quad p_Y(y) = \sum_x p(x, y). \tag{3}$$

Discrete Random Vectors

We can describe ***joint probability table*** by using a tabular. The marginal probabilities are described in the last row and the last column.

$y \backslash x$	x_1	x_2	$\cdots$	x_i	$\cdots$	$p_Y(y)$
y_1	$p(x_1, y_1)$	$p(x_2, y_1)$	$\cdots$	$p(x_i, y_1)$	$\cdots$	$p_Y(y_1)$
y_2	$p(x_1, y_2)$	$p(x_2, y_2)$	$\cdots$	$p(x_i, y_2)$	$\cdots$	$p_Y(y_2)$
$\vdots$	$\cdots$	$\cdots$	$\ddots$	$\cdots$	$\cdots$	$\vdots$
y_j	$p(x_1, y_2)$	$p(x_2, y_2)$	$\cdots$	$p(x_i, y_j)$	$\cdots$	$p_Y(y_j)$
$\vdots$	$\cdots$	$\cdots$	$\ddots$	$\cdots$	$\cdots$	$\vdots$
$p_X(x)$	$p_X(x_1)$	$p_X(x_2)$	$\cdots$	$p_X(x_i)$	$\cdots$	1

Table: The joint probability table of X and Y

Discrete Random Vectors

Example

A large insurance agency services a number of customers who have purchased both a homeowner's policy and an automobile policy from the agency. For each type of policy, a deductible amount must be specified. For an automobile policy, the choices are \$100 and \$250, whereas for a homeowner's policy, the choices are 0, \$100, and \$200. Suppose an individual with both types of policy is selected at random from the agency隆炉s files. Let X be the deductible amount on the automobile policy and Y be the deductible amount on the homeowner's policy. Suppose the joint p.f. is given in the accompanying joint probability table. Find the value of $P(Y \geqslant 100)$.

Discrete Random Vectors

x \ y	0	100	200
100	0.2	0.1	0.2
250	0.05	0.15	0.3

Solution. All possible (X, Y) pairs are respectively

$$(100, 0), (100, 100), (100, 200), (250, 0), (250, 100), \text{ and } (250, 200).$$

The joint p.f. specifies the probability associated with each one of these pairs, with any other pair having probability zero.

Then

$$\begin{aligned} p(100, 100) &= P(X = \$100 \text{ and } Y = \$100) \\ &= P(\$100 \text{ deductible on both policies}) = 0.1. \end{aligned}$$

Discrete Random Vectors

Solution. The probability $P(Y \geqslant 100)$ is computed by summing probabilities of all (x, y) pairs for which $y \geqslant 100$, that is,

$$\begin{aligned} P(Y \geqslant 100) &= \sum_{x} \sum_{y \geqslant 100} p(x, y) \\ &= p(100, 100) + p(250, 100) + p(100, 200) + p(250, 200) \\ &= 0.75. \end{aligned}$$

$\square$

Discrete Random Vectors

Example

Find $p_X(x)$ and $p_Y(y)$ in above Example.

Solution. The possible values of X are $x = 100$ and $x = 250$, so computing row totals in the joint probability table yields

$$p_X(100) = p(100, 0) + p(100, 100) + p(100, 200) = 0.5$$

and

$$p_X(250) = p(250, 0) + p(250, 100) + p(250, 200) = 0.5.$$

The marginal p.f. of X is then

$$p_X(x) = \begin{cases} 0.5 & \text{for } x = 100,\ 250, \\ 0 & \text{otherwise.} \end{cases}$$

Discrete Random Vectors

Similarly, the marginal p.f. of Y is obtained from column totals as

$$p_Y(y) = \begin{cases} 0.25 & \text{for } y = 0,\ 100, \\ 0.5 & \text{for } y = 200, \\ 0 & \text{otherwise.} \end{cases}$$

So $P(Y \geqslant 100) = p_Y(100) + p_Y(200) = 0.75$ as before. $\square$

Discrete Random Vectors

Example

Suppose that the joint probability table of X and Y is described in

$y \backslash x$	-1	0	2
0	0.1	0.3	a
1	0.1	0.1	b

Assume $P(XY = 0) = 0.7$. Find the values of a and b.

Solution. Since $P(XY = 0) = 0.7$,

$$\begin{aligned} P(XY = 0) &= P(\{X = 0\} \cup \{Y = 0\}) \\ &= P(X = 0) + P(Y = 0) - P(X = 0, Y = 0) \\ &= 0.4 + 0.4 + a - 0.3 = 0.7 \end{aligned}$$

Discrete Random Vectors

Thus $a = 0.2$. By the property $\sum_{(x,y)} p(x,y) = 1$, i.e., $a + b = 0.4$, we get $b = 0.2$. $\square$

Example

The distributions of X and Y are given in following Table.

x	-1	0	1
$p_X(x)$	$1/4$	$1/2$	$1/4$

y	0	1
$p_Y(y)$	$1/2$	$1/2$

Suppose that $P(XY = 0) = 1$. Find the joint probability function of X and Y.

Discrete Random Vectors

Solution. Since $P(XY = 0) = 1$, $p_X(0) = 1/2$, and $p_Y(0) = 1/2$,

$$P(XY = 0) = P(\{X = 0\} \cup \{Y = 0\})$$

$$= p_X(0) + p_Y(0) - p(0,0) = 1 - p(0,0) = 1.$$

Thus $p(0,0) = 0$, i.e., the joint probability table of X and Y is given in Table

$y \backslash x$	-1	0	1	$p_Y(y)$
0	$p(-1,0)$	0	$p(1,0)$	$1/2$
1	$p(-1,1)$	$p(0,1)$	$p(1,1)$	$1/2$
$p_X(x)$	$1/4$	$1/2$	$1/4$	1

.

Discrete Random Vectors

By using equation (3), we immediately obtain

$$\begin{aligned}
&0 + p(0,1) = 1/2,\\
&p(-1,1) + p(0,1) + p(1,1) = 1/2,\\
&p(-1,0) + p(-1,1) = 1/4,\\
&p(1,0) + p(1,1) = 1/4.
\end{aligned}$$

Since $p(x,y) \geqslant 0$, we obtain

$$p(0,1) = 1/2, \quad p(-1,1) = p(1,1) = 0, \quad p(-1,0) = p(1,0) = 1/4.$$

Thus

$y \backslash x$	-1	0	1	$p_Y(y)$
0	$1/4$	0	$1/4$	$1/2$
1	0	$1/2$	0	$1/2$
$p_X(x)$	$1/4$	$1/2$	$1/4$	1

$\square$

Discrete Random Vectors

We have known that the d.f. of a discrete random variable X is a step function, and all possible values are exactly those breakpoints of the step function.

How about the relationship between d.f. and p.f. of discrete bivariate random vectors (X, Y)?

Theoretically, we have equation (2), but it is really not that easy.

Discrete Random Vectors

Example

Suppose that the joint probability table of X and Y is Table 2. Find the joint d.f. of X and Y.

$y \backslash x$	0	1
0	1/2	0
1	0	1/2

Table: *The joint probability table of X and Y in Example 6*

Discrete Random Vectors

Solution. From the Fig. **??**, we observe that for those kind of points A, B and C in second, third and fourth quadrants, we have $F(x, y) = P(X \leqslant x, Y \leqslant y) = P(\emptyset) = 0$. For the kind of points D, E and F in first quadrant,

$$F(x, y) = P(X \leqslant x, Y \leqslant y) = P(X = 0, Y = 0) = 1/2.$$

For the point G, which is in the upper right corner of the point (1,1),

$$F(x, y) = P(X \leqslant x, Y \leqslant y) = P(\{X = 0, Y = 0\} \cup \{X = 1, Y = 1\}) = 1$$

So we have

$$F(x, y) = \begin{cases} 0 & \text{for } x < 0 \text{ or } y < 0, \\ 1/2 & \text{for } 0 \leqslant x < 1,\ y \geqslant 0 \text{ or } x \geqslant 1,\ 0 \leqslant y < 1, \\ 1 & \text{for } x \geqslant 1, y \geqslant 1. \end{cases}$$

$\square$

Section 3.1 The Distribution Function of a Random Variable

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Continuous Random Vectors

Let X and Y be two continuous random variables. The random variables X and Y are called ***(jointly) continuous*** if the joint distribution function $F(x, y)$ can be expressed by the ***joint probability density function*** (joint p.d.f.) $f(x, y)$ as

$$F(x, y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f(u, v) du dv.$$

The joint p.d.f. $f(x, y)$ has the following properties:

(i) $f(x, y) \geqslant 0$ for any point $(x, y) \in \mathbb{R} \times \mathbb{R}$;

(ii) $\displaystyle\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = 1.$

Continuous Random Vectors

Conversely, if a function $f(x, y)$ satisfies the two properties above, we can prove it must be a joint p.d.f. for some random vector (X, Y).
The probability $P[(X, Y) \in A]$ is obtained by integrating the joint p.d.f. over the domain A:

$$P[(X, Y) \in A] = \iint_{(x,y)\in A} f(x, y)dxdy.$$

Continuous Random Vectors

Example

A bank operates both a drive-up facility and a walk-up window. On a randomly selected day, let X be the proportion of time that the drive-up facility is in use (at least one customer is being served or waiting to be served) and Y be the proportion of time that the walk-up window is in use. Then the set of possible values for (X, Y) is the rectangle $D = \{(x, y) : 0 \leqslant x \leqslant 1, 0 \leqslant y \leqslant 1\}$. Suppose that the joint p.d.f. of (X, Y) is given by

$$f(x, y) = \begin{cases} \frac{6}{5}(x + y^2) & \text{for } 0 \leqslant x \leqslant 1, 0 \leqslant y \leqslant 1, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Verify that $f(x, y)$ is a legitimate joint p.d.f..
(b) Calculate the value of $P(0 \leqslant X \leqslant \frac{1}{4}, 0 \leqslant Y \leqslant \frac{1}{4})$.

Continuous Random Vectors

Solution. (a) To verify that this is a legitimate p.d.f., note that $f(x, y) \geqslant 0$ and

$$\begin{aligned}
&\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y)dxdy \\
&= \int_0^1\int_0^1 \frac{6}{5}(x+y^2)dxdy \\
&= \int_0^1\int_0^1 \frac{6}{5}xdxdy + \int_0^1\int_0^1 \frac{6}{5}y^2dxdy \\
&= \int_0^1 \frac{6}{5}xdx + \int_0^1 \frac{6}{5}y^2dy = \frac{6}{10} + \frac{6}{15} = 1.
\end{aligned}$$

Continuous Random Vectors

(b) The desired probability is

$$
\begin{aligned}
&P(0 \leqslant X \leqslant \frac{1}{4}, 0 \leqslant Y \leqslant \frac{1}{4}) \\
=&\int_0^{1/4}\int_0^{1/4} \frac{6}{5}(x+y^2)dxdy \\
=&\int_0^{1/4}\int_0^{1/4} \frac{6}{5}xdxdy + \int_0^{1/4}\int_0^{1/4} \frac{6}{5}y^2dxdy \\
=&\frac{7}{640} = 0.0109.
\end{aligned}
$$

$\square$

Continuous Random Vectors

Example

Let X and Y have density

$$f(x,y) = \begin{cases} 8xy & \text{if } 0 < y < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

What are $P(2X > 1, 2Y < 1)$ and $P(X + Y > 1)$? Find $F(x,y)$.

Solution. Notice that constraints $2X > 1$, $2Y < 1$ require that $(X,Y) \in A$, where A is the square with vertices $(1/2,0),(1,0),(1/2,1/2),(1,1/2)$ of Fig. 3.5(a). Hence,

$$P(2X > 1, 2Y < 1) = 8 \int_{\frac{1}{2}}^{1} \int_{0}^{\frac{1}{2}} f(x,y)dydx$$

$$= 8 \int_{\frac{1}{2}}^{1} xdx \int_{0}^{\frac{1}{2}} ydy = \frac{3}{8}.$$

Continuous Random Vectors

Likewise, $X + Y > 1$ if $(X, Y) \in B$, where B is the triangle with vertices $(1/2, 0), (1, 0), (1, 1)$ of Fig. 3.5(b). Hence,

$$P(X + Y > 1) = \iint\limits_{x+y>1,\ y<x} 8xydxdy = 8\int_{\frac{1}{2}}^{1} dx \int_{1-x}^{x} xydy = \frac{5}{6}.$$

Finally, for the points A, B and C satisfying $x \leqslant 0$ or $y \leqslant 0$, $F(x, y) = 0$.
For the point D satisfying $0 < y < x < 1$,

$$F(x, y) = \int_0^y dv \int_v^x 8uvdu = 2x^2y^2 - y^4.$$

Continuous Random Vectors

For the point E satisfying $0 < x < 1, y \geqslant x$,

$$F(x,y) = \int_0^x du \int_0^u 8uvdv = x^4.$$

For the point F satisfying $x \geqslant 1, 0 \leqslant y < 1$,

$$F(x,y) = \int_0^y dv \int_v^1 8uvdu = 2y^2 - y^4.$$

For the point (x,y) satisfying $x > 1, y \geqslant 1$, $F(x,y) = 1$. As a result,

$$F(x,y) = \begin{cases} 0 & \text{for } x \leqslant 0 \text{ or } y \leqslant 0, \\ 2x^2y^2 - y^4 & \text{for } 0 < x < y < 1, \\ x^4 & \text{for } 0 < x < 1, y \geqslant x, \\ 2y^2 - y^4 & \text{for } x \geqslant 1, 0 \leqslant y < 1, \\ 1 & \text{for } x > 1, y \geqslant 1. \end{cases}$$

$\square$

Continuous Random Vectors

For continuous random vector (X, Y), we have

$$F_X(x) = \int_{-\infty}^{x} \int_{-\infty}^{+\infty} f(u, v) dv du,$$

$$F_Y(y) = \int_{-\infty}^{y} \int_{-\infty}^{+\infty} f(u, v) du dv$$

since $F_X(x) = \lim_{y \to \infty} F(x, y)$ and $F_Y(y) = \lim_{x \to \infty} F(x, y)$.

By differentiating, we get

Definition

The ***marginal probability density functions*** *of X and Y are defined by*

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy \quad \text{and} \quad f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx. \tag{4}$$

Continuous Random Vectors

Example

Consider the joint p.d.f. $f(x, y)$ defined by

$$f(x, y) = cxye^{-x^2-y^2} \quad \text{for } x \geqslant 0, y \geqslant 0,$$

where $c > 0$ is a constant.
(a) Determine the constant c.
(b) Determine the joint d.f. of (X, Y).
(c) Determine the marginal density function of X.

Continuous Random Vectors

Solution. (a) Since $\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y)dxdy = 1$,

$$\int_0^{+\infty}\int_0^{+\infty} cxye^{-x^2-y^2}dxdy = c\int_0^{+\infty} xe^{-x^2}dx\int_0^{+\infty} ye^{-y^2}dy$$

$$=c\Big[-\frac{e^{-x^2}}{2}\Big|_0^{\infty}\Big]\Big[-\frac{e^{-y^2}}{2}\Big|_0^{\infty}\Big] = c(1/2)(1/2) = c/4.$$

Thus this function is a valid joint density fuction if and only if the constant c is equal to 4.

Continuous Random Vectors

(b) The joint distribution function of the pair (X, Y) is given by

$$\begin{aligned} F(x,y) &= \int_0^y \int_0^x 4uve^{-u^2-v^2}\,du\,dv \\ &= \int_0^x 2ue^{-u^2}\,du \int_0^y 2ve^{-v^2}\,dv \\ &= \Big(-e^{-u^2}\Big|_0^x \Big)\Big(-e^{-v^2}\Big|_0^y \Big) = (1-e^{-x^2})(1-e^{-y^2}) \end{aligned}$$

for $x \geqslant 0, y \geqslant 0$.

(c) The marginal density function of X is

$$\begin{aligned} f_X(x) &= \int_0^{+\infty} 4xye^{-x^2-y^2}\,dy = 2xe^{-x^2}\int_0^{+\infty} 2ye^{-y^2}\,dy \\ &= 2xe^{-x^2}\Big(-e^{-y^2}\Big|_0^{+\infty} \Big) = 2xe^{-x^2} \qquad \text{for } x \geqslant 0. \end{aligned}$$

$\square$

Continuous Random Vectors

Example

Suppose that (X, Y) are random vector that can only take values in the interval $[0, 2]$. Suppose that also that the joint d.f. of X and Y, for $0 \leqslant x \leqslant 2$ and $0 \leqslant y \leqslant 2$, as follows:

$$F(x, y) = \frac{1}{16} xy(x + y).$$

Determine the marginal p.d.f. $f_X(x)$ of just the random variable X.

Solution. Since (X, Y) takes value only in the interval $[0, 2]$, we know if either $x < 0$ or $y < 0$, then $F(x, y) = 0$. If both $x > 2$ and $y > 2$, then $F(x, y) = 1$. If $0 \leqslant x \leqslant 2$ and $y > 2$, then $F(x, y) = F(x, 2)$, and it means

$$F(x, y) = \frac{1}{8} x(x + 2).$$

Continuous Random Vectors

Similarly, if $0 \leqslant y \leqslant 2$ and $x > 2$, then

$$F(x, y) = \frac{1}{8} y(y + 2).$$

The function $F(x, y)$ has now been specified for every point in the xy-plane.
By letting $y \to \infty$, we find that the d.f. of the random variable X is

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0, \\ \frac{1}{8} x(x + 2) & \text{for } 0 \leqslant x \leqslant 2, \\ 1 & \text{for } x > 2. \end{cases}$$

By differentiating,

$$f_X(x) = \begin{cases} \frac{x + 1}{4} & \text{for } 0 \leqslant x \leqslant 2, \\ 0 & \text{otherwise.} \end{cases}$$

$\square$

Continuous Random Vectors

Example

A nut company markets cans of deluxe mixed nuts containing almonds, cashews, and peanuts. Suppose that the net weight of each can is exactly 1 1b, but the weight contribution of each type of nut is random. Because the three weights sum to 1, a joint probability model for any two gives all necessary information about the weight of the third type. Let X be the weight of almonds in a selected can and Y be the weight of cashews. Then the region of density is $D = \{(x, y) : 0 \leqslant x \leqslant 1, 0 \leqslant y \leqslant 1, x+y \leqslant 1\}$. Now let the joint p.d.f. for (X, Y) be

$$f(x, y) = \begin{cases} 24xy & \text{for } 0 \leqslant x \leqslant 1, 0 \leqslant y \leqslant 1, x + y \leqslant 1, \\ 0 & \text{otherwise.} \end{cases}$$

Compute the probability that the two types of nuts together make up at most 50% of the can.

Continuous Random Vectors

Solution. Let
$A = \{(x, y) : 0 \leqslant x \leqslant 0.5, 0 \leqslant y \leqslant 0.5, x + y \leqslant 0.5\}$.

$$P((X, Y) \in A) = \iint_A f(x, y)dxdy$$

$$= \int_0^{0.5} \int_0^{0.5-x} 24xydydx = 0.625.$$

□

Remark: For any fixed x, $f(x, y)$ increases with y; for fixed y, $f(x, y)$ increases with x. This is appropriate because the word deluxe implies that most of the can should consist of almonds and cashews rather than peanuts, so that the density function should be large near the upper boundary and small near the origin. The surface determined by $f(x, y)$ slopes upward from zero as (x, y) moves away from either axis.

Continuous Random Vectors

Let $F(x, y)$ be a joint distribution. Suppose that $\dfrac{\partial^2 F}{\partial x \partial y}$ exists and is nonnegative, except possibly on a finite collection of lines in $\mathbb{R}^2$. Then the function $f(x, y)$ can be obtained by

$$f(x, y) = \begin{cases} \dfrac{\partial^2 F}{\partial x \partial y} & \text{where this exists,} \\ 0 & \text{elsewhere.} \end{cases}$$

Continuous Random Vectors

Example

(*Uniform distribution*) Suppose that you pick a point (X, Y) at random in the rectangle $D = \{(x, y) : 0 < x < a, 0 < y < b\}$. That is, if A is a subset of D with area $|A|$, then we have

$$P((X, Y) \in A) = \frac{|A|}{ab}.$$

We call (X, Y) has uniform distribution on the domain D, denoted by $(X, Y) \sim U(D)$.
(a) Find the joint d.f. and p.d.f. of (X, Y).
(b) Find the marginal p.d.f. of X and Y respectively.

Continuous Random Vectors

Solution. (a) If $x \geqslant a, y \geqslant b$, then

$$F(x,y) = P(0 \leqslant X \leqslant a, 0 \leqslant Y \leqslant b) = \frac{ab}{ab} = 1.$$

If $0 \leqslant x \leqslant a,\ 0 \leqslant y \leqslant b$, then

$$F(x,y) = P(0 \leqslant X \leqslant x, 0 \leqslant Y \leqslant y) = \frac{xy}{ab}.$$

If $x \geqslant a,\ 0 \leqslant y \leqslant b$, then

$$F(x,y) = P(0 \leqslant X \leqslant a, 0 \leqslant Y \leqslant y) = \frac{ay}{ab} = \frac{y}{b}.$$

Similarly, if $0 \leqslant x \leqslant a,\ y \geqslant b$, then $F(x,y) = x/a$. Otherwise, $F(x,y) = 0$. That is,

Continuous Random Vectors

$$F(x,y) = \begin{cases} 1 & \text{for } x \geqslant a, y \geqslant b, \\ \dfrac{xy}{ab} & \text{for } 0 \leqslant x \leqslant a,\ 0 \leqslant y \leqslant b, \\ \dfrac{y}{b} & \text{for } x \geqslant a,\ 0 \leqslant y \leqslant b, \\ \dfrac{x}{a} & \text{for } 0 \leqslant x \leqslant a,\ y \geqslant b, \\ 0 & \text{elsewhere.} \end{cases}$$

Differentiating wherever possible gives the joint p.d.f. of (X, Y)

$$f(x,y) = \frac{\partial^2 F}{\partial x \partial y} = \begin{cases} \dfrac{1}{ab} & \text{for } 0 < x < a,\ 0 < y < b, \\ 0 & \text{otherwise.} \end{cases} \tag{5}$$

Continuous Random Vectors

(b) By equation (4), if $0 < x < a$, then

$$f_X(x) = \int_{-\infty}^{+\infty} f(x,y)dy = \int_0^b \frac{1}{ab}dy = \frac{1}{a}.$$

Otherwise, $f_X(x) = 0$. i.e., $X \sim U(0,a)$,

$$f_X(x) = \begin{cases} \dfrac{1}{a} & \text{for } 0 < x < a, \\ 0 & \text{otherwise.} \end{cases}$$

Similarly, $Y \sim U(0,b)$,

$$f_Y(y) = \begin{cases} \dfrac{1}{b} & \text{for } 0 < y < b, \\ 0 & \text{otherwise.} \end{cases}$$

$\square$

Continuous Random Vectors

In fact, the domain D in above example can be generalized to any finite region on $\mathbb{R}^2$. If $(X, Y) \sim U(D)$, then for any subregion $A \in D$,

$$P((X, Y) \in D) = \frac{|A|}{|D|},$$

where $|D|$ is the area of D. Obviously, the probability is only dependent on the subregion's area and independent of its position or its shape. The corresponding joint p.d.f. is

$$f(x, y) = \begin{cases} \dfrac{1}{|D|} & \text{if } (x, y) \in D, \\ 0 & \text{otherwise.} \end{cases} \tag{6}$$

Continuous Random Vectors

Example

If the joint p.d.f. of (X, Y) is specified by

$$f(x,y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(x-\mu_2)^2}{\sigma_2^2}\right]}, \tag{7}$$

$$-\infty < x < +\infty, \quad -\infty < y < +\infty,$$

where $-\infty < \mu_1 < +\infty$, $-\infty < \mu_2 < +\infty$, $\sigma_1 > 0$, $\sigma_2 > 0$, $-1 < \rho < 1$, then we say that (X, Y) has a ***bivariate normal distribution***, and denote it by $(X, Y) \sim N(\mu_1, \mu_2; \sigma_1^2, \sigma_2^2; \rho)$. Find the marginal p.d.f. of X and Y.

Solution. Let $t = \rho\dfrac{x-\mu_1}{\sigma_1\sqrt{1-\rho^2}} - \dfrac{y-\mu_2}{\sigma_2\sqrt{1-\rho^2}}$. Then

$$-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(x-\mu_2)^2}{\sigma_2^2}\right] = -\frac{t^2}{2} - \frac{(x-\mu_1)^2}{2\sigma_1^2}.$$

Continuous Random Vectors

By equation (4),

$$
\begin{aligned}
f_X(x) &= \int_{-\infty}^{+\infty} f(x,y)dy \\
&= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \int_{-\infty}^{+\infty} \exp\Big\{ -\frac{1}{2}\Big[\rho\frac{x-\mu_1}{\sigma_1\sqrt{1-\rho^2}} - \frac{y-\mu_2}{\sigma_2\sqrt{1-\rho^2}}\Big]^2 - \frac{(x-\mu_1)^2}{2\sigma_1^2}\Big\}dy \\
&= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \int_{-\infty}^{+\infty} \exp\Big\{ -\frac{t^2}{2} - \frac{(x-\mu_1)^2}{2\sigma_1^2}\Big\} \cdot \big(-\sigma_2\sqrt{1-\rho^2}\big)dt \\
&= \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(x-\mu_1)^2}{2\sigma_1}} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{t^2}{2}} dt \\
&= \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(x-\mu_1)^2}{2\sigma_1}}.
\end{aligned}
$$

It is exactly a normal distribution with parameters μ_1 and σ_1^2, i.e., $X \sim N(\mu_1, \sigma_1^2)$. Similarly, we can prove $Y \sim N(\mu_2, \sigma_2^2)$. $\square$

Continuous Random Vectors

Let $\Theta \sim U(-\pi, \pi)$, and put $X = cos\Theta$ and $Y = sin\Theta$. As shown in Exercise 2.45 in Chapter 2, X and Y are both arcsine random variables, each having density $(1/\pi)\sqrt{1-x^2}$ for $-1 < x < 1$. Next, since $X^2 + Y^2 = 1$, the pair (X, Y) takes values only on the unit circle

$$C := \{(x; y) : x^2 + y^2 = 1\}.$$

Thus, $P((X, Y) \in C) = 1$. On the other hand, if X and Y have a joint density $f(x, y)$, then

$$P((X, Y) \in C) = \iint_C f(x, y)dxdy = 0$$

because a double integral over a set of zero area must be zero. So, if X and Y have a joint density, this would imply that $1 = 0$. Since this is not true, there can be no joint density.

Section 3.2 Independence of Random Variables

Given the joint probability (density) function of X and Y, we could get the marginal probability (density) functions of X and Y. However, to be given the marginal distributions does not in general uniquely determine a joint distribution. In this section, we will see only if X and Y are independent, the joint distribution function could be obtained by the marginal distributions.

Independence of Random Variables

Example

Let X and Y have joint probability function given by

$$p(0,0) = \frac{1}{6},\ p(0,1) = \frac{1}{3},\ p(1,0) = \frac{1}{12},\ p(1,1) = \frac{5}{12},$$

and let U and V have joint probability function given by

$$p(0,0) = \frac{1}{4},\ p(0,1) = \frac{1}{4},\ p(1,0) = 0,\ p(1,1) = \frac{1}{2}.$$

Then, summing to get the marginal probability functions shows that:

$$p_X(0) = p_U(0) = \frac{1}{2},\ p_X(1) = p_U(1) = \frac{1}{2};$$

and

$$p_Y(0) = p_V(0) = \frac{1}{4},\ p_Y(1) = p_V(1) = \frac{3}{4}.$$

SAME m.p.f., DIFFERENT j.p.f.. □

Independence of Random Variables

In Chapter 2, we pointed out that one way of defining independence of two events A and B is via the condition $P(A \cap B) = P(A) \cdot P(B)$. Here is an analogous definition for the independence of two random variables. Intuitively, if and only if for any subsets I_1, $I_2 \in \mathbb{R}$,

$$P(X \in I_1, Y \in I_2) = P(X \in I_1)P(Y \in I_2) \tag{8}$$

holds, we may obtain the independence of X and Y. However,

Independence of Random Variables

Definition

Two ***random variables*** X ***and*** Y ***are said to be independent*** *if for every pair of* x *and* y,

$$F(x, y) = F_X(x)F_Y(y). \tag{9}$$

We denote it by $X \perp Y$.

Independence of Random Variables

Obviously, equation (9) can be deduced by equation (8). In fact, the two equations are equivalent. Here let us just see an example. Let $I_1 = (x_1, x_2]$ and $I_2 = (y_1, y_2]$. We will prove that if equation (9) holds, then for any $x_1 < x_2$ and $y_1 < y_2$,

$$\begin{aligned}
& P(x_1 < X \leqslant x_2,\ y_1 < Y \leqslant y_2) \\
= {} & F(x_2, y_2) - F(x_2, y_1) - F(x_1, y_2) + F(x_1, y_1) \\
= {} & F(x_2)F(y_2) - F(x_2)F(y_1) - F(x_1)F(y_2) + F(x_1)F(y_1) \\
= {} & [F(x_2) - F(x_1)] \cdot [F(y_2) - F(y_1)] \\
= {} & P(x_1 < X \leqslant x_2)P(y_1 < Y \leqslant y_2).
\end{aligned}$$

Independence of Random Variables

In fact, the following definition of independence of two random variables is equivalent to Definition 2, which is more useful in many applications.

Definition

Two ***random variables*** X ***and*** Y ***are said to be independent*** *if for every pair of* x *and* y,

$$\begin{cases} p(x,y) = p_X(x) \cdot p_Y(y) & \textit{when } X \textit{ and } Y \textit{ are discrete} \\ \qquad\qquad \textit{or} & \\ f(x,y) = f_X(x) \cdot f_Y(y) & \textit{when } X \textit{ and } Y \textit{ are continuous.} \end{cases} \tag{10}$$

If (10) is not satisfied for all (x,y), *then* X *and* Y *are said to be dependent.*

Independence of Random Variables

It is easy to get equation (10) from equation (9). Next we prove that equation (10) implies equation (9) for the discrete case. For any x and y,

$$\begin{aligned} P(X \leqslant x, Y \leqslant y) &= \sum_{x_i \leqslant x,\ y_j \leqslant y} P(X = x_i, Y = y_j) \\ &= \sum_{x_i \leqslant x,\ y_j \leqslant y} P(X = x_i) P(Y = y_j) \\ &= \sum_{x_i \leqslant x} P(X = x_i) \sum_{y_j \leqslant y} P(Y = y_j) \\ &= P(X \leqslant x, Y \leqslant y). \end{aligned}$$

The continuous case can be proved similarly.

Independence of Random Variables

Example

Suppose that $X \perp Y$.

x	-2	-1	0	$1/2$
$p_X(x)$	$1/4$	$1/3$	$1/12$	$1/3$

y	$-1/2$	1	3
$p_Y(y)$	$1/2$	$1/4$	$1/4$

Find the joint p.f. of (X, Y).

Independence of Random Variables

Solution. Since $X \perp Y$,

$$P(X=-2, Y=-1/2) = P(X=-2)P(Y=-1/2) = \frac{1}{4} \times \frac{1}{2} = \frac{1}{8}.$$

Similarly, we can obtain the joint probability table of X and Y in Table

$y \backslash x$	-2	-1	0	$1/2$	$p_Y(y)$
$-1/2$	$1/8$	$1/6$	$1/24$	$1/6$	$1/2$
1	$1/16$	$1/12$	$1/48$	$1/12$	$1/4$
3	$1/16$	$1/12$	$1/48$	$1/12$	$1/4$
$p_X(x)$	$1/4$	$1/3$	$1/12$	$1/3$	1

$\square$

Independence of Random Variables

Example

Suppose that the joint probability table of X and Y is in Table.

$y \backslash x$	1	2	3
1	$1/6$	$1/9$	$1/8$
2	$1/3$	$1/a$	$1/b$

Find the values of a and b such that $X \perp Y$.

Independence of Random Variables

Solution. For every pair of x and y, we need

$$p(x, y) = p_X(x) \cdot p_Y(y).$$

Select two simple equalities

$$p(2, 1) = p_X(2)p_Y(1) \text{ and } p(3, 1) = p_X(3)p_Y(1).$$

That is,

$$\frac{1}{9} = \frac{1}{3}\Big(\frac{1}{9} + \frac{1}{a}\Big), \quad \frac{1}{18} = \frac{1}{3}\Big(\frac{1}{18} + \frac{1}{b}\Big).$$

Thus we get $a = 9/2, b = 9$. $\square$

Independence of Random Variables

Example

Suppose that n machines produce components, and all the machines produce the same number c of components. We pick one at random and let its serial number be (X, Y), where X is the machine number and Y is the component index. Are X and Y independent?

Solution. Since the joint p.d.f. of (X, Y) is

$$f(x, y) = (nc)^{-1}, \quad 1 \leqslant x \leqslant n,\ 1 \leqslant y \leqslant c$$

and

$$f_X(x) = \frac{1}{n},\ 1 \leqslant x \leqslant n, \quad \text{and } f_Y(y) = \frac{1}{c},\ 1 \leqslant y \leqslant c.$$

Obviously, $f(x, y) = f_X(x) f_Y(y)$ for all (x, y). So X and Y are independent in this case. $\square$

Independence of Random Variables

Example

A man and a woman decide to meet at a certain location. If each of them independently arrives at a time uniformly distributed between 12 noon and 1 P.M., then find the probability that the first to arrive has to wait no longer than 10 minutes.

Solution. If we let X and Y denote, respectively, the time that the man and the woman arrive, then X and Y are independent random variables, each of which is uniformly distributed over (0, 60).

Independence of Random Variables

The desired probability, $P(|X - Y| \leqslant 10)$, i.e., $1 - [P(X + 10 < Y) + P(Y + 10 < X)]$, which, by symmetry, equals $1 - 2P(X + 10 < Y)$, is obtained as follows:

$$\begin{aligned}
P(|X - Y| \leqslant 10) &= 1 - \ 2P(X + 10 < Y) \\
&= 1 - 2 \iint_{x+10<y} f(x, y) dx dy \\
&= 1 - 2 \iint_{x+10<y} f_X(x) f_Y(y) dx dy \\
&= 1 - 2 \int_{10}^{60} dy \int_{0}^{y-10} \left(\frac{1}{60}\right)^2 dx \\
&= 1 - \frac{2}{(60)^2} \int_{10}^{60} (y - 10) dy = \frac{11}{36}.
\end{aligned}$$

$\square$

Independence of Random Variables

Example

Let us consider the Buffon's needle experiment. In terms of random variables the experiment just discussed can be phrased as: we denote by X the distance from the center of the needle to the nearest line and by Θ the acute angle between the needle and the lines. We assume that the random variables X and Θ are independent. X is uniform in the interval $(0, d/2)$. and Θ is uniform in the interval $(0, \pi)$.

Independence of Random Variables

From this it follows that

$$f(x,\theta) = f_X(x) \cdot f_\Theta(\theta) = \frac{2}{d}\frac{1}{\pi}, \qquad 0 \leqslant x \leqslant d/2,\ 0 \leqslant \theta \leqslant \pi$$

and 0 elsewhere.

The needle intersects the lines if $x \leqslant \dfrac{l \sin\theta}{2}$. Hence

$$P(X \leqslant \frac{l\sin\theta}{2}) = \frac{2}{\pi d}\int_0^\pi \frac{l\sin\theta}{2} d\theta = \frac{2l}{\pi d}.$$

$\square$

Independence of Random Variables

Example

Suppose that $(X, Y) \sim N(\mu_1, \mu_2; \sigma_1^2, \sigma_2^2; \rho)$. Prove $X \perp Y$ if and only if $\rho = 0$.

Solution. If $\rho = 0$, then the independence of X and Y is obvious. Conversely, since $f(x, y) = f_X(x) f_Y(y)$ for any x and y,

$$\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(x-\mu_2)^2}{\sigma_2^2}\right]}$$

$$= \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(x-\mu_1)^2}{2\sigma_1}} \frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{(y-\mu_2)^2}{2\sigma_2}}.$$

Because x and y are arbitrary, we take $x = \mu_1$, $y = \mu_2$. Then

$$\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} = \frac{1}{\sqrt{2\pi}\sigma_1} \frac{1}{\sqrt{2\pi}\sigma_2},$$

i.e., $\rho = 0$. $\square$

Independence of Random Variables

Example

Are X and Y independent?
(a) Suppose that (X, Y) has a uniform distribution over $D = \{(x, y) : 0 < x < a, 0 < y < b\}$.
(b) Suppose that (X, Y) has a uniform distribution over the unit circle.

Solution. (a) By using the result in Example 12, we know for any $x, y \in D$, $f(x, y) = f_X(x) f_Y(y)$. Thus X and Y are independent.

Independence of Random Variables

(b) The joint p.d.f. of (X, Y) is

$$f(x,y) = \begin{cases} \dfrac{1}{\pi} & \text{for } x^2 + y^2 = 1, \\ 0 & \text{otherwise.} \end{cases}$$

The marginal p.d.f. of X and Y are

$$f_X(x) = \int_{-\infty}^{+\infty} f(x,y)dy = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi} dy = \frac{2}{\pi}\sqrt{1-x^2}, \quad |x| < 1,$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x,y)dx = \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \frac{1}{\pi} dx = \frac{2}{\pi}\sqrt{1-y^2}, \quad |y| < 1.$$

Since $f(x,y) \neq f_X(x)f_Y(y)$, X and Y are dependent. $\square$

Independence of Random Variables

In fact, we have a necessary and sufficient condition for the random variables X and Y to be independent in a rectangular.

Theorem

Let a, b, c and d be given values such that $-\infty \leqslant a < b \leqslant +\infty$ and $-\infty \leqslant c < d \leqslant +\infty$, and let S be a rectangle in the xy-plane:

$$S = \{(x, y) \mid a \leqslant x \leqslant b \text{ and } c \leqslant y \leqslant d\}.$$

Suppose that $f(x, y) = 0$ for every point (x, y) outside S. Then the continuous (discrete) random variables X and Y are independent if and only if their joint p.d.f. (or p.f.) can be expressed as

$$f(x, y) = h(x)g(y)$$

for all points in S.

Independence of Random Variables

Proof. Suppose that the joint p.d.f. of (X, Y) is

$$f(x,y) = \begin{cases} h(x)g(y) & \text{if } a \leqslant x \leqslant b,\ c \leqslant y \leqslant d, \\ 0 & \text{otherwise.} \end{cases}$$

If $a \leqslant x \leqslant b$, then,

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)dy = \int_c^d h(x)g(y)dy = h(x)\int_c^d g(y)dy.$$

Similarly, if $c \leqslant y \leqslant d$, then

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)dx = \int_a^b h(x)g(y)dx = g(y)\int_c^d h(x)dx.$$

Since $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)dxdy = 1$, $\int_a^b h(x)dx \int_c^d g(y)dy = 1$. So for every pair (x,y), we have $f(x,y) = f_X(x)f_Y(y)$. Thus $X \perp Y$.

The proof of the other part is obvious. $\square$

Independence of Random Variables

Example

Suppose that the joint p.d.f. of X and Y is

$$f(x, y) = 6e^{-2x}e^{-3y}, \qquad 0 < x < \infty,\ 0 < y < \infty$$

and is equal to 0 outside this region. Are the random variables independent?

Solution. The joint density function factors, and thus the random variables, are independent (with one being exponential with rate 2 and the other exponential with rate 3). $\square$

Independence of Random Variables

If the joint density function is

$$f(x, y) = 24xy, \quad 0 < x < 1,\ 0 < y < 1,\ 0 < x + y < 1$$

and is equal to 0 otherwise, are the random variables independent?

Theorem

Let X and Y be two independent random variables. Then for any real functions $g(\cdot)$ and $h(\cdot)$, $g(X)$ and $h(Y)$ are independent.

Summary

Theorem

Two ***random variables*** X ***and*** Y ***are independent*** *if and only if for every pair of* x *and* y,

$$\begin{cases} p(x,y) = p_X(x) \cdot p_Y(y) & \text{when } X \text{ and } Y \text{ are discrete} \\ \qquad\qquad \text{or} & \\ f(x,y) = f_X(x) \cdot f_Y(y) & \text{when } X \text{ and } Y \text{ are continuous.} \end{cases} \tag{11}$$

Theorem

Let X *and* Y *be two independent random variables. Then for any real functions* $g(\cdot)$ *and* $h(\cdot)$, $g(X)$ *and* $h(Y)$ *are independent.*

Section 3.3 Conditional Distributions

For two random variables X and Y, we naturally consider the problem of conditional probability, such as, calculate the values of
(i) $P\Big(\{\text{the weight} \geqslant 70kg\}\big|\{\text{the height} \geqslant 175cm\}\Big)$ or
(ii) $P\Big(\{\text{the weight} \geqslant 70kg\}\big|\{\text{the height} = 175cm\}\Big)$.
For question (i),

$$P\Big(\{\text{the weight } \geqslant 70kg\}\big|\{\text{the height } \geqslant 175cm\}\Big)$$
$$= \frac{P\Big(\{\text{the weight } \geqslant 70kg\} \cap \{\text{the height } \geqslant 175cm\}\Big)}{P(\text{the height } \geqslant 175cm)}$$

But for question (ii), if we suppose that the distribution of the height is continuous, the probability $P(\text{the height} = 175cm)$ is 0, then the conditional probability is meaningless. So we must discuss the problem in different cases: discrete case and continuous case.

Section 3.3 Conditional Distributions

- 3.3.1 Discrete Case
- 3.3.2 Continuous Case

Discrete Case

If X and Y are discrete random variables, then it is natural to define the ***conditional probability function of*** X ***given that*** $Y = y$, by

$$p_{X|Y}(x|y) = P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{p(x, y)}{p_Y((y)}$$

for all values of y such that $p_Y(y) > 0$. Similarly, the ***conditional probability function of*** Y ***given that*** $X = x$ is given by

$$p_{Y|X}(y|x) = P(Y = y|X = x) = \frac{P(X = x, Y = y)}{P(X = x)} = \frac{p(x, y)}{p_X(x)}$$

for all values of x such that $p_X(x) > 0$.

Discrete Case

Example

Suppose that $p(x, y)$, the joint probability function of X and Y, is given by $p(0,0) = 0.4$, $p(0,1) = 0.2$, $p(1,0) = 0.1$, $p(1,1) = 0.3$. Calculate the conditional probability function of X given that $Y = 1$.

Solution. We first note that

$$p_Y(1) = \sum_x p(x, 1) = p(0,1) + p(1,1) = 0.5.$$

Hence,

$$p_{X|Y}(0|1) = \frac{p(0,1)}{p_Y(1)} = \frac{2}{5}$$

and

$$p_{X|Y}(1|1) = \frac{p(1,1)}{p_Y(1)} = \frac{3}{5}.$$

□

Discrete Case

Example

Suppose that X is a random variable, which can be selected randomly from the numbers 1, 2, 3 and 4. Let Y be an another random variable, which can be selected randomly from positive integers 1 to X. Find the joint probability function of (X, Y).

Solution. It is obvious that all possible values of X and Y are 1, 2, 3 and 4. So we have 4×4 different pairs of (X, Y). Then

$$P(X = 1, Y = 1) = P(X = 1) \cdot P(Y = 1|X = 1) = \frac{1}{4} \cdot 1 = \frac{1}{4}.$$

$$P(X = 2, Y = 1) = P(X = 2) \cdot P(Y = 1|X = 2) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}.$$

$$P(X = 3, Y = 1) = P(X = 3) \cdot P(Y = 1|X = 3) = \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{12}.$$

$$P(X = 4, Y = 1) = P(X = 4) \cdot P(Y = 1|X = 4) = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}.$$

Discrete Case

$y \backslash x$	1	2	3	4
1	$1/4$	$1/4 \cdot 1/2$	$1/4 \cdot 1/3$	$1/4 \cdot 1/4$
2	0	$1/4 \cdot 1/2$	$1/4 \cdot 1/3$	$1/4 \cdot 1/4$
3	0	0	$1/4 \cdot 1/3$	$1/4 \cdot 1/4$
4	0	0	0	$1/4 \cdot 1/4$

$\square$

Discrete Case

Example

A store has a certain goods. Let X be the number of customers entering this store in a specified period of time. Suppose that $X \sim P(\lambda)$ and the probability of the event that each customer purchases the certain goods is p. If customs are independent, then find the probability function of the number of customers who purchase the certain goods.

Solution. Let Y be the number of customers who purchase the certain goods. Since $X \sim P(\lambda)$,

$$P(X = m) = \frac{\lambda^m}{m!} e^{-\lambda}, \quad m = 0, 1, 2, \cdots .$$

Discrete Case

Given that $X = m$, i.e., the number of customers entering this store is m, the conditional probability function of Y is binomial distribution, i.e.,

$$P(Y = k|X = m) = \binom{m}{k} p^k (1-p)^{m-k}, \quad k = 0, 1, 2, \cdots, m.$$

Using total probability formula, we have

$$\begin{aligned}
P(Y = k) &= \sum_{m=k}^{+\infty} P(X = m) P(Y = k|X = m) \\
&= \sum_{m=k}^{+\infty} \frac{\lambda^m}{m!} e^{-\lambda} \binom{m}{k} p^k (1-p)^{m-k} \\
&= e^{-\lambda} \frac{\lambda^k}{k!} \sum_{m=k}^{+\infty} \frac{[(1-p)\lambda]^{m-k}}{(m-k)!} \\
&= e^{-\lambda} \frac{\lambda^k}{k!} e^{\lambda(1-p)} = \frac{\lambda^k}{k!} e^{-\lambda p}, \quad k = 0, 1, 2, \cdots
\end{aligned}$$

Section 3.3 Conditional Distributions

- 3.3.1 Discrete Case
- 3.3.2 Continuous Case

Continuous Case

Definition

If X and Y have joint p.d.f. $f(x,y)$, then ***the conditional probability density function of X given $Y = y$*** *is given by*

$$f_{X|Y}(x|y) = \begin{cases} \dfrac{f(x,y)}{f_Y(y)} & if\ 0 < f_Y(y) < +\infty, \\ 0 & elsewhere. \end{cases} \tag{12}$$

Similarly, the ***conditional probability density function of Y given $X = x$*** *is given by*

$$f_{Y|X}(y|x) = \begin{cases} \dfrac{f(x,y)}{f_X(x)} & if\ 0 < f_X(x) < +\infty, \\ 0 & elsewhere. \end{cases} \tag{13}$$

Continuous Case

In fact, for each fixed value of y such that $f_Y(y) > 0$, the function $f_{X|Y}(x|y)$ will be a p.d.f. for X over the real line since $f_{X|Y}(x|y) \geqslant 0$ and $\int_{-\infty}^{+\infty} f_{X|Y}(x|y)dx = 1$. Similarly, for each fixed value of x such that $f_X(x) > 0$, the function $f_{Y|X}(y|x)$ is also a p.d.f. for Y over the real line.

Continuous Case

Remark

A conditional p.d.f. is not the result of conditional on a set of probability zero. Actually, the value of $f_{X|Y}(x|y)$ *is a limit:*

$$f_{X|Y}(x|y) = \lim_{\varepsilon \to 0} \frac{\partial}{\partial x} P(X \leqslant x | y - \varepsilon < Y \leqslant y + \varepsilon). \tag{14}$$

The conditioning event $\{y - \varepsilon < Y \leqslant y + \varepsilon\}$ *in equation (14) has positive probability if the marginal p.d.f. of* Y *is positive at* y*. Or Multiply the left side by* dx *and the right side by* $(dxdy)/dy$ *to equation (12) to get*

Continuous Case

$$
\begin{aligned}
f_{X|Y}(x|y)dx &= \frac{f(x,y)dxdy}{f_Y(y)dy} \\
&\approx \frac{P(x \leqslant X \leqslant x+dx, y \leqslant Y \leqslant y+dy)}{P(y \leqslant Y \leqslant y+dy)} \\
&= P(x \leqslant X \leqslant x+dx \mid y \leqslant Y \leqslant y+dy).
\end{aligned}
$$

In other words, for small values dx and dy, $f_{X|Y}(x|y)dx$ is approximately the conditional probability that X is between x and $x+dx$ given that Y is between y and $y+dy$.

Continuous Case

Example

Given

$$f(x,y) = \begin{cases} k & \text{for } 0 < x < y < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Determine $f_{X|Y}(x|y)$ and $f_{Y|X}(y|x)$.

Solution. The joint p.d.f. is given to be a constant in the region $\{(x,y) : 0 < x < y < 1\}$. This gives

$$\iint_{\mathbb{R}^2} f(x,y)dxdy = \int_0^1 \int_0^y kdxdy = \int_0^1 kydy = \frac{k}{2} = 1 \Rightarrow k = 2.$$

$$f_X(x) = \int_{-\infty}^{+\infty} f(x,y)dy = \int_x^1 kdy = k(1-x), \qquad 0 < x < 1,$$

Continuous Case

and

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x,y)dx = \int_0^y kdx = ky, \qquad 0 < y < 1.$$

From equations (12) and (13), we get if $0 < y < 1$, then

$$f_{X|Y}(x|y) = \frac{f(x,y)}{f_Y(y)} = \begin{cases} \dfrac{1}{y} & \text{for } 0 < x < y, \\ 0 & \text{otherwise.} \end{cases}$$

and if $0 < x < 1$, then

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \begin{cases} \dfrac{1}{1-x} & \text{for } x < y < 1, \\ 0 & \text{otherwise.} \end{cases}$$

$\square$

Continuous Case

Example

Given

$$f(x,y) = \begin{cases} \dfrac{3}{16}(4-2x-y) & \text{for } x>0,\ y>0,\ 2x+y<4, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Determine $f_{Y|X}(y|x)$.
(b) Determine the value of $P(Y \geqslant 2 \mid X \leqslant 1/2)$.
(c) Determine the value of $P(Y \geqslant 2 \mid X = 1/2)$.

Continuous Case

Solution. (a) First let us find the marginal p.d.f. $f_X(x)$. If $0 < x < 2$, then

$$f_X(x) = \int_{-\infty}^{+\infty} f(x,y)dy = \int_0^{4-2x} \frac{3}{16}(4-2x-y)dy = \frac{3}{8}(2-x)^2.$$

If $x \leqslant 0$ or $x \geqslant 1$, then $f(x,y) = 0$. i.e.,

$$f_X(x) = \begin{cases} \frac{3}{8}(2-x)^2 & \text{for } 0 < x < 2, \\ 0 & \text{otherwise,} \end{cases}$$

So when $0 < x < 2$, we have

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \begin{cases} \frac{4-2x-y}{2(2-x)^2} & \text{for } 0 < y < 4-2x, \\ 0 & \text{otherwise.} \end{cases}$$

Continuous Case

(b) According to the definition of conditional probability, we have

$$
\begin{aligned}
P(Y \geqslant 2 \mid X \leqslant 1/2) &= \frac{P(X \leqslant 1/2, Y \geqslant 2)}{P(X \leqslant 1/2)} \\
&= \frac{\displaystyle\int_0^{1/2} dx \int_2^{4-2x} \frac{3}{16}(4-2x-y)dy}{\displaystyle\int_0^{1/2} \frac{3}{8}(2-x)^2 dx} \\
&= \frac{\displaystyle\int_0^{1/2} (\frac{3}{8} - \frac{3}{4}x + \frac{3}{8}x^2)dx}{\displaystyle\int_0^{1/2} \frac{3}{8}(2-x)^2 dx} = \frac{7}{64}.
\end{aligned}
$$

Continuous Case

(c) Since

$$f_{Y|X}\left(y\middle|\frac{1}{2}\right) = f_{Y|X}(y|x)\Big|_{x=1/2} = \begin{cases} \dfrac{2(3-y)}{9} & \text{for } 0 < y < 3, \\ 0 & \text{otherwise,} \end{cases}$$

we get

$$\begin{aligned} P(Y \geqslant 2 \mid X = 1/2) &= \int_2^{+\infty} f_{Y|X}\left(y\middle|\frac{1}{2}\right) dy \\ &= \int_2^3 \frac{2(3-y)}{9} dy = \frac{1}{9}. \end{aligned}$$

$\square$

Continuous Case

Example

Suppose that (X, Y) has a uniform distribution over the unit circle. Find $f_{Y|X}(y|x)$.

Solution. We have known if $|x| < 1$ then

$$f_X(x) = \frac{2}{\pi}\sqrt{1 - x^2}.$$

Otherwise, $f_X(x) = 0$. So when $|x| < 1$,

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \begin{cases} \dfrac{1}{2\sqrt{1 - x^2}} & \text{for } -\sqrt{1 - x^2} \leqslant y \leqslant \sqrt{1 - x^2}, \\ 0 & \text{otherwise.} \end{cases}$$

$\square$

Continuous Case

If we take $x = 1/2$, then $Y \sim U(-\sqrt{3}/2, \sqrt{3}/2)$. If we take $x = 1/3$, then $Y \sim U(-\frac{2\sqrt{2}}{3}, \frac{2\sqrt{2}}{3})$. We can easily find the distribution of Y is dependent on the value of X.

Continuous Case

Example

Suppose that $(X, Y) \sim N(\mu_1, \mu_2; \sigma_1^2, \sigma_2^2; \rho)$. Find $f_{Y|X}(y|x)$ and $f_{X|Y}(x|y)$.

Solution. Given $X = x$, the conditional p.d.f. of Y is

$$f_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)} = \frac{1}{\sigma_2\sqrt{2\pi(1-\rho^2)}} \exp\Big\{ -\frac{1}{2(1-\rho^2)\sigma_2^2}\Big(y - \mu_2 - \rho\sigma_2\frac{x-\mu_1}{\sigma_1}\Big)$$

This is also a normal distribution

$$N\Big(\mu_2 + \rho\sigma_2\frac{x-\mu_1}{\sigma_1}, (1-\rho^2)\sigma_2^2\Big).$$

Here x is a constant. Similarly, given $Y = y$, the conditional p.d.f. of X is a normal distribution $N\Big(\mu_1 + \rho\sigma_1\frac{y-\mu_2}{\sigma_2}, (1-\rho^2)\sigma_1^2\Big)$. $\square$

Continuous Case

For two dimension normal distribution, we can see not only the marginal distributions are normal, but the conditional distributions are also normal.

Summary

- **Discrete**

$$p_{X|Y}(x|y) = P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{p(x, y)}{p_Y((y)}$$

$$p_{Y|X}(y|x) = P(Y = y|X = x) = \frac{P(X = x, Y = y)}{P(X = x)} = \frac{p(x, y)}{p_X(x)}$$

- **Continuous**

$$f_{X|Y}(x|y) = \begin{cases} \dfrac{f(x, y)}{f_Y(y)} & if\ 0 < f_Y(y) < +\infty, \\ 0 & elsewhere. \end{cases}$$

$$f_{Y|X}(y|x) = \begin{cases} \dfrac{f(x, y)}{f_X(x)} & if\ 0 < f_X(x) < +\infty, \\ 0 & elsewhere. \end{cases}$$

Section 3.4 One Function of Two Random Variables

Given two random variables X and Y and a function $\varphi(x, y)$, we form a new random variable Z as

$$Z = \varphi(X, Y).$$

Given the joint p.d.f. $f(x, y)$ of X and Y, how does one obtain the p.d.f. $f_Z(z)$ of Z? Problems of this type are of interest from a practical standpoint. For example, a received signal in a communication scene usually consists of the desired signal buried in noise, and this formulation in that case reduces to $Z = X + Y$. It is important to know the statistics of the incoming signal for proper receiver design. In this context, we shall analyze problems of the type.

Section 3.4 One Function of Two Random Variables

- 3.4.1 Discrete Case
- 3.4.2 Continuous Case

Discrete Case

There are two important things for a random variable X: all possible values taken by X and the corresponding probabilities. So to find the probability function of the random variable $Z = \varphi(X, Y)$, we should firstly make sure what values Z can take according to all possible values of (X, Y), then calculate the corresponding probability for each possible value of Z by using the joint p.f. $p(x, y)$ of (X, Y).

Discrete Case

Example

Assume the joint p.f. of random variable (X, Y) is given by Table

$y \backslash x$	0	1	2	3	4	5
0	0	0.01	0.03	0.05	0.07	0.09
1	0.01	0.02	0.04	0.05	0.06	0.08
2	0.01	0.03	0.05	0.05	0.05	0.06
3	0.01	0.02	0.04	0.06	0.06	0.05

Determine the probability functions of (a) $X+Y$, (b) $\min(X, Y)$ and (c) $\max(X, Y)$.

Discrete Case

Solution. (a) The possible outcomes of $X+Y$ are: $0, 1, 2, 3, 4, 5, 6, 7, 8$. Then

$$
\begin{aligned}
P(X+Y=0) &= P(X=0, Y=0) = 0, \\
P(X+Y=1) &= P(\{X=0, Y=1\} \cup \{X=1, Y=0\}) \\
&= P(X=0, Y=1) + P(X=1, Y=0) \\
&= 0.01 + 0.01 = 0.02, \\
P(X+Y=2) &= P(\{X=0, Y=2\} \cup \{X=1, Y=1\} \cup \{X=2, Y= \\
&= P(X=0, Y=2) + P(X=1, Y=1) + P(X=2, Y = \\
&= 0.03 + 0.02. + 0.01 = 0.06.
\end{aligned}
$$

Similarly, we can calculate other probabilities.

$X+Y$	0	1	2	3	4	5	6	7	8
P	0	0.02	0.06	0.13	0.19	0.24	0.19	0.12	0.05

Discrete Case

(b) The possible outcomes of $\min(X, Y)$ are: $0, 1, 2, 3$. Then

$$
\begin{aligned}
&P(\min(X,Y)=0)\\
&= P\Big(\{X=0,Y=3\}\cup\{X=0,Y=2\}\cup\{X=0,Y=1\}\\
&\quad\cup\{X=0,Y=0\}\cup\{X=1,Y=0\}\cup\{X=2,Y=0\}\\
&\quad\cup\{X=3,Y=0\}\cup\{X=4,Y=0\}\cup\{X=5,Y=0\}\Big)\\
&= P(X=0,Y=3)+P(X=0,Y=2)+P(X=0,Y=1)\\
&\quad+P(X=0,Y=0)+P(X=1,Y=0)+P(X=2,Y=0)\\
&\quad+P(X=3,Y=0)+P(X=4,Y=0)+P(X=5,Y=0)\\
&= 0.01+0.01+0.01+0+0.01+0.03+0.05+0.07+0.09=0.28.
\end{aligned}
$$

Other probabilities can be calculated similarly.

$\min(X,Y)$	0	1	2	3
P	0.28	0.30	0.25	0.17

Discrete Case

(c) The possible outcomes of $\max(X, Y)$ are: $0, 1, 2, 3, 4, 5$. Then

$$P(\max(X, Y) = 0) = P(\{X = 0, Y = 0\}) = 0.$$

$$\begin{aligned}
&P(\max(X, Y) = 1) \\
&= P(\{X = 1, Y = 0\} \cup \{X = 0, Y = 1\} \cup \{X = 1, Y = 1\}) \\
&= P(X = 1, Y = 0) + P(X = 1, Y = 1) + P(X = 1, Y = 0) \\
&= 0.01 + 0.02 + 0.01 = 0.04.
\end{aligned}$$

Other probabilities can be calculated similarly.

$\max(X, Y)$	0	1	2	3	4	5
P	0	0.04	0.16	0.28	0.24	0.28

$\square$

Discrete Case

Example

Independent random variables X and Y take the values 1 or -1 respectively. Suppose $P(X = 1) = a$ and $P(Y = 1) = b$. A third random variable Z is defined by $Z = cos\dfrac{\pi(X+Y)}{2}$. Find the p.f. of Z.

Solution. All possible values of Z are 1 and -1, and

$$P(Z = 1) = P(X + Y = 0) = a(1 - b) + b(1 - a).$$

Similarly, $P(Z = -1) = ab + (1 - b)(1 - a)$.

$\square$

Discrete Case

It is often important to be able to calculate the distribution of $X + Y$ from the distributions of X and Y when X and Y are independent.

Theorem

If X and Y are independent discrete random variables, then $X + Y$ has probability function

$$p_{X+Y}(n) = \sum_k p_X(k) p_Y(n-k). \tag{15}$$

The function p_{X+Y} is called the convolution of p_X and p_Y, and is written as

$$p_{X+Y} = p_X * p_Y.$$

Discrete Case

Example

Let X and Y be independent geometric variables with common probability function

$$p(k) = p(1-p)^{k-1}, \qquad k = 1, 2, \cdots$$

Determine the probability function of $X + Y$.

Solution. Because the event $\{X + Y = n\}$ may be written as the union of the disjoint events $\{X = k, Y = n - k\}$, $0 \leqslant k \leqslant n$, we have

Discrete Case

$$
\begin{aligned}
P(X+Y=n) &= \sum_{k=1}^{n-1} P(X=k, Y=n-k) \\
&= \sum_{k=1}^{n-1} P(X=k)P(Y=n-k) \\
&= \sum_{k=1}^{n-1} p(1-p)^{k-1} p(1-p)^{n-k-1} \\
&= (n-1)p^2(1-p)^{n-2}, \qquad n = 2, 3, \cdots .
\end{aligned}
$$

$\square$

Discrete Case

Example

Assume X and Y are independent, and $X \sim B(n_1, p)$, $Y \sim B(n_2, p)$. Prove $X + Y \sim B(n_1 + n_2, p)$.

Proof.

$$
\begin{aligned}
P(X+Y=k) &= \sum_{k_1} P(X=k_1)P(Y=k-k_1) \\
&= \sum_{k_1=0}^{k} \binom{n_1}{k_1} p^{k_1}(1-p)^{n_1-k_1} \cdot \binom{n_2}{k-k_1} p^{k-k_1}(1-p)^{n_2-k+k_1} \\
&= p^k(1-p)^{n_1+n_2-k} \sum_{k_1=0}^{k} \binom{n_1}{k_1} \cdot \binom{n_2}{k-k_1} \\
&= \binom{n_1+n_2}{k} p^k(1-p)^{n_1+n_2-k}, \qquad k = 0, 1, 2, \cdots, n_1+n_2.
\end{aligned}
$$

Thus $X + Y \sim B(n_1 + n_2, p)$. $\square$

Discrete Case

Recalling the interpretation of a binomial random variable, and without any computation at all, we can immediately conclude that $X + Y$ is binomial with parameters $(n_1 + n_2, p)$. This follows because X represents the number of successes in n_1 independent trials, each of which results in a success with probability p; similarly, Y represents the number of successes in n_2 independent trials, each of which results in a success with probability p. Hence, given that X and Y are assumed independent, it follows that $X + Y$ represents the number of successes in $n_1 + n_2$ independent trials when each trial has a probability p of resulting in a success. Therefore, $X + Y$ is a binomial random variable with parameters $(n_1 + n_2, p)$.

Discrete Case

Example

Assume X and Y are independent, and $X \sim P(\lambda)$, $Y \sim P(\mu)$. Prove $X + Y \sim P(\lambda + \mu)$.

Solution. $X + Y$ has the following probability function:

$$
\begin{aligned}
P(X+Y=n) &= \sum_{k=0}^{n} P(X=k, Y=n-k) = \sum_{k=0}^{n} P(X=k)P(Y=n-k) \\
&= \sum_{k=0}^{n} e^{-\lambda}\frac{\lambda^k}{k!}e^{-\mu}\frac{\mu^{n-k}}{(n-k)!} = e^{-(\lambda+\mu)}\sum_{k=0}^{n}\frac{\lambda^k\mu^{n-k}}{k!(n-k)!} \\
&= \frac{e^{-(\lambda+\mu)}}{n!}\sum_{k=0}^{n}\frac{n!}{k!(n-k)!}\lambda^k\mu^{n-k} \\
&= \frac{e^{-(\lambda+\mu)}}{n!}(\lambda+\mu)^n
\end{aligned}
$$

Thus, $X + Y$ has a Poisson distribution with parameter $\lambda + \mu$. $\square$

Section 3.4 One Function of Two Random Variables

- 3.4.1 Discrete Case
- 3.4.2 Continuous Case

Continuous Case

Let X and Y be random variables having joint p.d.f. $f(x, y)$. Let Z be given by $Z = \varphi(X, Y)$, where φ is a real-valued function whose domain contains the range of X and Y. In order to determine the p.d.f. of Z, we need to find the d.f. of Z first. For a fixed $z \in \mathbb{R}$, the event $\{Z \leqslant z\}$ is equivalent to the event $\{(X, Y) \in A_z\}$, where A_z is the subset of $\mathbb{R}^2$ defined by

$$A_z = \{(x, y) | \varphi(x, y) \leqslant z\}.$$

Continuous Case

Thus

$$\begin{aligned} F_Z(z) =& P(Z \leqslant z) = P(\varphi(X,Y) \leqslant z) \\ =& P((X,Y) \in A_z) = \iint\limits_{A_z} f(x,y)dxdy. \end{aligned}$$

If we can find a nonnegative function g such that

$$\iint\limits_{A_z} f(x,y)dxdy = \int_{-\infty}^{z} g(v)dv, \qquad -\infty < z < \infty,$$

then g is necessary a density of Z.

In this section, we mainly concern the cases when $\varphi(X,Y) = X + Y$, $\min(X,Y)$ and $\max(X,Y)$.

Continuous Case

1. The case of $\boldsymbol{X+Y}$

Set $Z = X + Y$. Then

$$A_z = \{(x, y) | x + y \leqslant z\}$$

is just the half-plane to the lower left of the line $x + y = z$. Thus

$$F_Z(z) = \iint\limits_{A_z} f(x, y) dx dy = \int_{-\infty}^{\infty} \Big(\int_{-\infty}^{z-x} f(x, y) dy \Big) dx.$$

Make the change of variable $y = v - x$ in the inner integral. Then

Continuous Case

$$
\begin{aligned}
F_Z(z) &= \int_{-\infty}^{\infty} \Big(\int_{-\infty}^{z} f(x, v - x) dv \Big) dx \\
&= \int_{-\infty}^{z} \Big(\int_{-\infty}^{\infty} f(x, v - x) dx \Big) dv,
\end{aligned}
$$

where we have interchanged the order of integration. Thus the density of $Z = X + Y$ is given by

$$
f_{X+Y}(z) = \int_{-\infty}^{\infty} f(x, z - x) dx, \qquad -\infty < z < \infty. \tag{16}
$$

If X and Y are independent, then equation (16) can be rewritten as

$$
f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx, \qquad -\infty < z < \infty. \tag{17}
$$

Continuous Case

That means the density of the sum of two independent random variables is the convolution of the individual densities. Equation (17) can be written as

$$f_{X+Y} = f_X * f_Y.$$

If X and Y are nonnegative independent random variables, then $f_{X+Y}(z) = 0$ for $z \leqslant 0$ and

$$f_{X+Y}(z) = \int_0^z f_X(x) f_Y(z-x) dx, \qquad 0 < z < \infty. \tag{18}$$

Continuous Case

Example

Let X and Y be two independent random variables, uniformly distributed in the same interval $[0, 1]$. Compute the distribution of $X + Y$, and compare with the distribution of $2X$.

Solution. The density of X is

$$f_X(x) = \begin{cases} 1 & \text{if } x \in (0, 1), \\ 0 & \text{otherwise.} \end{cases}$$

The density of Y is the same. Thus $f_{X+Y}(z) = 0$ for $z \leqslant 0$. For $z > 0$,

$$f_X(x) f_Y(z - x) = \begin{cases} 1 & \text{if } 0 \leqslant x \leqslant 1,\ 0 \leqslant z - x \leqslant 1, \\ 0 & \text{otherwise.} \end{cases}$$

Continuous Case

If $0 \leqslant z \leqslant 1$, then

$$f_{X+Y}(z) = \int_0^z f_X(x) f_Y(z-x) dx = \int_0^z 1 dx = z.$$

If $1 < z \leqslant 2$, then

$$f_{X+Y}(z) = \int_0^z f_X(x) f_Y(z-x) dx = \int_{z-1}^1 1 dx = 2 - z.$$

If $2 < z < \infty$, then

$$f_{X+Y}(z) = \int_0^z f_X(x) f_Y(z-x) dx = \int_0^z 0 dx = 0.$$

Continuous Case

In summary,

$$f_{X+Y}(z) = \begin{cases} z & \text{if } 0 \leqslant z \leqslant 1, \\ 2 - z & \text{if } 1 < z \leqslant 2, \\ 0 & \text{elsewhere.} \end{cases}$$

But, obviously,

$$f_{2X}(x) = \begin{cases} 1/2 & \text{if } x \in (0, 1), \\ 0 & \text{otherwise.} \end{cases}$$

It is because X and Y are independent, but X and X are not. $\square$

Continuous Case

Example

Let X and Y be independent random variables, which have exponential distributions with parameter λ_1 and λ_2 respectively. Find the distribution of $X+Y$.

Solution. Let the distributions of X and Y be respectively

$$f_X(x) = \begin{cases} \lambda_1 e^{-\lambda_1 x} & \text{for } x \geqslant 0, \\ 0 & \text{for } x < 0, \end{cases} \text{ and } f_Y(y) = \begin{cases} \lambda_2 e^{-\lambda_2 y} & \text{for } y \geqslant 0, \\ 0 & \text{for } y < 0. \end{cases}$$

By using equation (18), we have $f_{X+Y}(z) = 0$ for $z \leqslant 0$ and for $z > 0$,

$$\begin{aligned} f_{X+Y}(z) &= \int_0^z \lambda_1 e^{-\lambda_1 x} \lambda_2 e^{-\lambda_2 (z-x)} dx \\ &= \lambda_1 \lambda_2 e^{-\lambda_2 z} \int_0^z e^{(-\lambda_1+\lambda_2)x} dx \\ &= \frac{\lambda_1 \lambda_2}{\lambda_2 - \lambda_1} (e^{-\lambda_1 z} - e^{-\lambda_2 z}). \end{aligned}$$

$\square$

Continuous Case

Example

Let X and Y be independent random variables having the respective normal densities $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$. Prove $X + Y$ has the normal distribution $N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Proof. Let $X_1 = X - \mu_1$ and $Y_1 = Y - \mu_2$. Then

$$X + Y = X_1 + Y_1 + \mu_1 + \mu_2.$$

By using Corollary 2.5.7,

$$X_1 \sim N(0, \sigma_1^2) \text{ and } Y_1 \sim N(0, \sigma_2^2).$$

Continuous Case

Firstly, we will show $X_1 + Y_1 \sim N(0, \sigma_1^2 + \sigma_2^2)$. Since

$$f_{X_1}(x_1) = \frac{1}{\sqrt{2\pi}\ \sigma_1} e^{-x_1^2/2\sigma_1^2}, \qquad -\infty < x_1 < \infty$$

and

$$f_{Y_1}(y_1) = \frac{1}{\sqrt{2\pi}\ \sigma_2} e^{-y_1^2/2\sigma_2^2}, \qquad -\infty < y_1 < \infty,$$

we get

$$\begin{aligned} f_{X_1+Y_1}(z) &= \int_{-\infty}^{\infty} f_{X_1}(x_1) f_Y(z - x_1) dx_1 \\ &= \frac{1}{2\pi\sigma_1\sigma_2} \int_{-\infty}^{\infty} \exp\Big[-\frac{1}{2}\Big(\frac{x_1^2}{\sigma_1^2} + \frac{(z - x_1)^2}{\sigma_2^2}\Big)\Big] dx_1 \\ &= \frac{1}{2\pi\sigma_1\sigma_2} \exp\Big\{ -\frac{z^2}{2(\sigma_1^2 + \sigma_2^2)}\Big\} \int_{-\infty}^{\infty} \exp\Big\{ -\frac{\sigma_1^2 + \sigma_2^2}{2\sigma_1^2\sigma_2^2}\Big(x_1 - \frac{z\sigma_1^2}{\sigma_1^2 + \sigma_2^2}\Big)^2\Big\} \end{aligned}$$

Continuous Case

Let $t = \dfrac{\sqrt{\sigma_1^2+\sigma_2^2}}{\sigma_1\sigma_2}\Big(x_1 - \dfrac{z\sigma_1^2}{\sigma_1^2+\sigma_2^2}\Big)$. Then

$$\begin{aligned} f_{X_1+Y_1}(z) &= \frac{1}{2\pi\sigma_1\sigma_2}\exp\Big\{-\frac{z^2}{2(\sigma_1^2+\sigma_2^2)}\Big\}\frac{\sigma_1\sigma_2}{\sqrt{\sigma_1^2+\sigma_2^2}}\int_{-\infty}^{\infty} e^{-t^2/2}dt \\ &= \frac{1}{\sqrt{2\pi\sigma_1^2+\sigma_2^2}}\ \exp\Big\{-\frac{z^2}{2(\sigma_1^2+\sigma_2^2)}\Big\}. \end{aligned}$$

i.e., $X_1 + Y_1 \sim N(0, \sigma_1^2+\sigma_2^2)$. By using Corollary 2.5.7 again,

$$X + Y \sim N(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2).$$

$\square$

Continuous Case

Corralary

If $X \sim N(\mu_1, \sigma_1^2)$, $Y \sim (\mu_2, \sigma_2^2)$ *and* $X \perp Y$, *then*

$$aX + bY + c \sim N(a\mu_1 + b\mu_2 + c, a^2\sigma_1^2 + b^2\sigma_2^2),$$

where a, b *are constants.*

Continuous Case

Example

Data from the National Oceanic and Atmospheric Administration indicate that the yearly precipitation in Los Angeles is a normal random variable with a mean of 12.08 inches and a standard deviation of 3.1 inches. Assume that the precipitation totals for the next 2 years are independent.

(a) Find the probability that the total precipitation during the next 2 years will exceed 25 inches.

(b) Find the probability that next year隆炉s precipitation will exceed that of the following year by more than 3 inches.

Continuous Case

Solution. Let X_1 and X_2 be the precipitation totals for the next 2 years. Then X_1 and X_2 have the same normal distribution $N(12.08, 3.1^2)$.

$X_1 + X_2 \sim N(24.16, 2 \cdot 3.1^2)$. $2 \cdot 3.1^2 = 19.22$. It follows that

$$\begin{aligned} P(X_1 + X_2 > 25) =& P\Big(\frac{X_1 + X_2 - 24.16}{\sqrt{19.22}} > \frac{25 - 24.16}{\sqrt{19.22}}\Big) \\ =& P(Z > 0.1916) \approx 0.4240 \end{aligned}$$

Continuous Case

(b) By using Corollary 3.4.2, we get $X_1 - X_2$ is normal with mean 0 and variance 19.22. Hence,

$$\begin{aligned} P(X_1 > X_2 + 3) =& P(X_1 - X_2 > 3) \\ =& P\Big(\frac{X_1 - X_2}{\sqrt{19.22}} > \frac{3}{\sqrt{19.22}}\Big) \\ =& P(Z > 0.6843) \approx 0.2469. \end{aligned}$$

Thus there is a 42.4% chance that the total precipitation in Los Angeles during the next 2 years will exceed 25 inches, and there is a 24.69% chance that next year隆炉s precipitation will exceed that of the following year by more than 3 inches. □

Continuous Case

By using similar ways of finding the p.d.f. of $X+Y$, we can obtain the probability density functions of $X-Y$, $X \cdot Y$ and X/Y.

(1) Let $Z = X - Y$. Then the p.d.f. of Z is

$$f_Z(z) = \int_{-\infty}^{+\infty} f(x, x-z)dx = \int_{-\infty}^{+\infty} f(z+y, y)dy.$$

If X and Y are independent, then

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(x) f_Y(x-z)dx = \int_{-\infty}^{+\infty} f_X(z+y) f_Y(y)dy.$$

Continuous Case

(2) Let $Z = X \cdot Y$. Then

$$f_Z(z) = \int_{-\infty}^{+\infty} f(x, z/x) \frac{1}{|x|} dx = \int_{-\infty}^{+\infty} f(z/y, y) \frac{1}{|y|} dy.$$

If X and Y are independent, then

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(x) f_Y(z/x) \frac{1}{|x|} dx = \int_{-\infty}^{+\infty} f_X(z/y) f_Y(y) \frac{1}{|y|} dy.$$

Continuous Case

(3) Let $Z = X/Y$. Then

$$f_Z(z) = \int_{-\infty}^{+\infty} f(x, x/z)\frac{|x|}{z^2}dx = \int_{-\infty}^{+\infty} f(yz, y)|y|dy.$$

If X and Y are independent, then

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(x)f_Y(x/z)\frac{|x|}{z^2}dx = \int_{-\infty}^{+\infty} f_X(yz)f_Y(y)|y|dy.$$

Continuous Case

2. The case of $\max(X, Y)$

Set

$$Z = \max(X, Y) = \begin{cases} X & \text{for } X > Y, \\ Y & \text{for } X \leqslant Y. \end{cases}$$

We have

$$\begin{aligned} F_Z(z) &= P(\max(X, Y) \leqslant z) \\ &= P(\{X \leqslant z, X > Y\} \cup \{Y \leqslant z, X \leqslant Y\}) \\ &= P(X \leqslant z, X > Y) + P(Y \leqslant z, X \leqslant Y). \end{aligned}$$

Continuous Case

Since $\{X > Y\}$ and $\{X \leqslant Y\}$ are mutually exclusive sets that form a partition,

$$F_Z(z) = P(X \leqslant z, Y \leqslant z) = F(z, z).$$

If X and Y are independent, then

$$F_Z(z) = F_X(z)F_Y(z) \tag{19}$$

and hence

$$f_Z(z) = F_X(z)f_Y(z) + f_X(z)F_Y(z). \tag{20}$$

Continuous Case

If X and Y are independent and have identical distribution function F and probability density function f, then equation (19) becomes

$$F_Z(z) = [F(z)]^2. \tag{21}$$

Equation (20) becomes

$$f_Z(z) = 2F(z)f(z). \tag{22}$$

Continuous Case

3. The case of $\min(X, Y)$

Set

$$W = \min(X, Y) = \begin{cases} Y & \text{for } X > Y, \\ X & \text{for } X \leqslant Y. \end{cases}$$

Thus,

$$\begin{aligned} F_W(w) &= P(\min(X, Y) \leqslant w) \\ &= P(\{Y \leqslant w, X > Y\} \cup \{X \leqslant w, X \leqslant Y\}). \end{aligned}$$

Continuous Case

Since the event $\{\min(X, Y) \leqslant w\}$ contains many cases, we consider its complement. Thus

$$\begin{aligned} F_W(w) &= 1 - P(\min(X, Y) > w) = 1 - P(X > w, Y > w) \\ &= F_X(w) + F_Y(w) - F_{X,Y}(w, w). \end{aligned}$$

If X and Y are independent, then

$$F_W(w) = 1 - P(X > w)P(Y > w) = \ 1 - [1 - F_X(w)][1 - F_Y(w)]. \tag{23}$$

Continuous Case

If X and Y are independent and have the same distribution function F and probability density function f, then

$$F_W(w) = 1 - [1 - F(w)]^2.$$

And

$$f_W(w) = F_W^{'}(w) = 2[1 - F(w)]f(w). \tag{24}$$

Continuous Case

Example

Suppose that $X_1 \sim Exp(\alpha)$, $X_2 \sim Exp(\beta)$ and $X_1 \perp X_2$. Let $Z = \max(X, Y)$ and $W = \min(X, Y)$. Determine the distributions of Z and W.

Solution. Since $X_1 \sim Exp(\alpha)$ and $X_2 \sim Exp(\beta)$,

$$F_X(x) = \begin{cases} 1 - e^{-\alpha x} & \text{for } x > 0, \\ 0 & \text{for } x \leqslant 0 \end{cases} \text{ and } F_Y(y) = \begin{cases} 1 - e^{-\beta y} & \text{for } y > 0, \\ 0 & \text{for } y \leqslant 0. \end{cases}$$

By using equation (19), we get

$$F_Z(z) = F_X(z)F_Y(z) = \begin{cases} (1 - e^{-\alpha z})(1 - e^{-\beta z}) & \text{for } z > 0, \\ 0 & \text{for } z \leqslant 0 \end{cases}$$

Continuous Case

and hence

$$f_Z(z) = \begin{cases} \alpha e^{-\alpha z} + \beta e^{-\beta z} - (\alpha+\beta)e^{-(\alpha+\beta)z} & \text{for } z > 0, \\ 0 & \text{for } z \leqslant 0. \end{cases}$$

By using equation (23), we can obtain

$$F_W(w) = 1 - [1 - F_X(w)][1 - F_Y(w)] \begin{cases} 1 - e^{-(\alpha+\beta)w} & \text{for } w > 0, \\ 0 & \text{for } w \leqslant 0, \end{cases}$$

and hence

$$f_W(w) = \begin{cases} (\alpha+\beta)e^{-(\alpha+\beta)w} & \text{for } w > 0, \\ 0 & \text{for } w \leqslant 0. \end{cases}$$

i.e., $W \sim Exp(\alpha+\beta)$. $\square$

Continuous Case

The distribution of max and min have many applications. For example, a married couple that holds a joint life with last survivor annuity receives a fixed amount of income each month until both spouses have died. Consider the case of a married couple currently ages x and y with respective future lifetimes T_x and T_y. For two lives's future lifetime $T_{xy} = \max(T_x, T_y)$, we have

$$F_{T_{xy}}(t) = P(\max(T_x, T_y) \leqslant t).$$

Section 3.5 Transformation of Two Random Variables

In the spirit of the previous section, let us look at an immediate generalization. Suppose that X and Y are two random variables with joint p.d.f. $f(x, y)$. Given two functions $g_1(x, y)$ and $g_2(x, y)$, define two new random variables

$$\begin{cases} Z_1 = g_1(X, Y), \\ Z_2 = g_2(X, Y). \end{cases}$$

How does one determine their joint p.d.f. of (Z_1, Z_2) by the joint p.d.f. of (X, Y)?

We consider only the case when the system possesses a unique solution: $x = h_1(z_1, z_2)$ and $y = h_2(z_1, z_2)$. The following proposition can then be proved.

Transformation of Two Random Variables

Property

Let (X, Y) be a continuous random vector and its joint p.d.f. is $f_{X,Y}$, $(x, y) \in D$. Let $Z_1 = g_1(X, Y)$ and $Z_2 = g_2(X, Y)$.

Suppose that the functions $x = h_1(z_1, z_2)$ and $y = h_2(z_1, z_2)$ have continuous partial derivatives (with respect to z_1 and z_2) for all (z_1, z_2) and that the Jacobian of the transformation

$$J(z_1, z_2) := \frac{\partial(h_1, h_2)}{\partial(z_1, z_2)} = \begin{vmatrix} \dfrac{\partial h_1}{\partial z_1} & \dfrac{\partial h_1}{\partial z_2} \\ \dfrac{\partial h_2}{\partial z_1} & \dfrac{\partial h_2}{\partial z_2} \end{vmatrix}$$

is not identical to zero.

Transformation of Two Random Variables

Property

Let

$$G = \{(z_1, z_2)|\ z_1 = g_1(x, y), z_2 = g_2(x, y), (x, y) \in D\}.$$

Then, we can write that

$$f_{Z_1,Z_2}(z_1, z_2) = \begin{cases} f_{X,Y}(h_1(z_1, z_2), h_2(z_1, z_2))|J(z_1, z_2)| \\ \qquad\qquad\qquad\qquad\qquad\qquad \text{for } (z_1, z_2) \in G, \\ 0 \qquad\qquad\qquad\qquad\qquad\qquad \text{otherwise.} \end{cases} \tag{25}$$

Transformation of Two Random Variables

Example

Let X_1 and X_2 be jointly continuous random variables with probability density function f_{X_1,X_2}. Let $Y_1 = X_1 + X_2$ and $Y_2 = X_1 - X_2$. Find the joint density function of Y_1 and Y_2 in terms of f_{X_1,X_2}.

Solution. Let $g_1(x_1, x_2) = x_1 + x_2$ and $g_2(x_1, x_2) = x_1 - x_2$. Then

$$J(x_1, x_2) := \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = 2.$$

Furthermore, since the equations $y_1 = x_1 + x_2$ and $y_2 = x_1 - x_2$ have $x_1 = (y_1 + y_2)/2$, $x_2 = (y_1 - y_2)/2$ as their solution, it follows from equation (25) that the desired density is

$$f_{Y_1,Y_2}(y_1, y_2) = \frac{1}{2} f_{X_1,X_2}\left(\frac{y_1 + y_2}{2}, \frac{y_1 - y_2}{2}\right).$$

$\square$

Transformation of Two Random Variables

For instance, if X_1 and X_2 are independent uniform (0, 1) random variables, then

$$f_{Y_1,Y_2}(y_1,y_2) = \begin{cases} \dfrac{1}{2} & \text{for } 0 \leqslant y_1 + y_2 \leqslant 2,\ 0 \leqslant y_1 - y_2 \leqslant 2, \\ 0 & \text{otherwise.} \end{cases}$$

Or if X_1 and X_2 are independent exponential random variables with respective parameters λ_1 and λ_2, then

$$f_{Y_1,Y_2}(y_1,y_2) = \begin{cases} \frac{\lambda_1\lambda_2}{2} \exp\left\{ -\lambda_1 \frac{y_1+y_2}{2} - \lambda_2 \frac{y_1-y_2}{2}\right)\right\} \\ \qquad\qquad \text{for } 0 \leqslant y_1 + y_2 \leqslant 2,\ 0 \leqslant y_1 - y_2 \leqslant 2, \\ 0 \qquad \text{otherwise.} \end{cases}$$

Transformation of Two Random Variables

Finally, if X_1 and X_2 are independent standard normal random variables, then

$$\begin{aligned} f_{Y_1,Y_2}(y_1,y_2) &= \frac{1}{4\pi} e^{-[(y_1+y_2)^2/8+(y_1-y_2)^2/8]} \\ &= \frac{1}{4\pi} e^{-(y_1^2+y_2^2)/4} \\ &= \frac{1}{\sqrt{4\pi}} e^{-y_1^2/4} \cdot \frac{1}{\sqrt{4\pi}} e^{-y_2^2/4}. \end{aligned}$$

Thus, not only do we obtain that both $X_1 + X_2$ and $X_1 - X_2$ are normal with mean 0 and variance 2, but we also conclude that these two random variables are independent.

Transformation of Two Random Variables

Even if we only want the distribution of one function of (X, Y), Proposition 2 still works. Suppose that we wish to obtain the density function of the transformation $Z = g_1(X, Y)$ of the continuous random vector (X, Y). We can define an auxiliary variable $W = g_2(X, Y)$ satisfying the system in Proposition.

$$f_Z(z) = \int_{-\infty}^{\infty} f_{Z,W}(z, w)dw$$

$$= \int_{-\infty}^{\infty} f_{X,Y}(h_1(z, w), h_2(z, w))|J(z, w)|dw$$

and

$$f_W(w) = \int_{-\infty}^{\infty} f_{Z,W}(z, w)dz$$

$$= \int_{-\infty}^{\infty} f_{X,Y}(h_1(z, w), h_2(z, w))|J(z, w)|dz.$$

Transformation of Two Random Variables

Remark

(i) *Generally, we choose a very simple auxiliary variable* W*, for example,* $W = X$.
(ii) *In the particular case when* $g_1(x, y)$ *is a linear transformation of* x *and* y*, it suffices to choose another linear transformation of* x *and* y*; for the partial derivatives of the functions* h_1 *and* h_2 *to be continuous. Indeed, these partial derivatives are constants, and therefore, they are continuous at any point* (z, w).

Transformation of Two Random Variables

Example

Let $X \sim U(0,1)$ and $Y \sim U(0,1)$ be two independent random variables and let $Z = X + Y$. Find the distribution of Z.

Solution. To obtain the density function of Z, we define the auxiliary variable $W = X$. Then, the system

$$z = x + y,$$
$$w = x$$

has the unique solution $x = w$, $y = z - w$. Moreover, the partial derivatives of the functions $h_1(z,w) = w$ and $h_2(z,w) = z - w$ are continuous for any (z,w) and the Jacobian

$$J(z,w) = \begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = -1$$

is not zero for all (z,w).

Transformation of Two Random Variables

Consequently, we can write that

$$f_{Z,W}(z,w) = f_{X,Y}(w, z-w) \cdot |-1| = f_X(w) f_Y(z-w).$$

It follows that

$$f_{Z,W}(z,w) = 1 \cdot 1 = 1 \quad \text{if } 0 < w < 1 \text{ and } 0 < z - w < 1.$$

If $0 < z < 1$, then the set of possible values of w is the interval $(0, z)$. If $1 \leqslant z < 2$, then the set of possible values of w is the interval $(z-1, 1)$. Thus

$$f_Z(z) = \begin{cases} \displaystyle\int_0^z 1 dw = z & \text{if } 0 < z < 1, \\ \displaystyle\int_{z-1}^1 1 dw = 2 - z & \text{if } 1 \leqslant z < 2, \\ 0 & \text{elsewhere.} \end{cases}$$

$\square$

Section 3.6 Numerical Characteristics of Random Vectors

For one random variable X, we have the expected value $\mu = E(X)$ and the variance $\sigma^2 = Var(X)$.

How to describe the expected value and the variance for a random vector (X, Y)?

Intuitively, let $\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$. Then,

$$\mu = E(\mathbf{X}) = \begin{pmatrix} E(X_1) \\ E(X_2) \end{pmatrix}, \qquad \Sigma = Var(\mathbf{X}) = \begin{pmatrix} Var(X_1) \\ Var(X_2) \end{pmatrix}.$$

Is it reasonable?

The answer is **NO**, because we do not consider the relationship between X and Y.

Numerical Characteristics of Random Vectors

In this section, we will introduce the covariance and a scaled version called the correlation between two random variables, which give us information about the relationship between the random variables. For convenience, we first introduce the expectation of sums and products, which is of independent interesting and important.

Section 3.6 Numerical Characteristics of Random Vectors

- 3.6.1 Expectation of Sums and Products
- 3.6.2 Covariance and Correlation

Expectation of Sums and Products

Theorem

Let X and Y be jointly distributed random variables with p.f. $p(x,y)$ or p.d.f. $f(x,y)$ according to whether the variables are discrete or continuous. Then the expected value of a function $h(X,Y)$, denoted by $E[h(X,Y)]$, is given by

$$E[h(X,Y)] = \begin{cases} \sum_x \sum_y h(x,y) \cdot p(x,y) & if\ (X,Y)\ are\ discr \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x,y) \cdot f(x,y) dxdy & if\ \ (X,Y)\ are\ cont \end{cases}$$

The expectation $E[h(X,Y)]$ exists if and only if

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |h(x,y)| \cdot f(x,y) dxdy < \infty.$$

Expectation of Sums and Products

Example

An accident occurs at a point X that is uniformly distributed on a road of length L. At the time of the accident, an ambulance is at a location Y that is also uniformly distributed on the road. Assuming that X and Y are independent, find the expected distance between the ambulance and the point of the accident.

Solution. We need to compute $E(|X - Y|)$. Since the joint density function of X and Y is

$$f(x, y) = \frac{1}{L^2}, \quad 0 < x < L,\ 0 < y < L,$$

it follows from the Proposition that

$$E(|X - Y|) = \frac{1}{L^2} \int_0^L \int_0^L |x - y| dy dx.$$

Expectation of Sums and Products

Now,

$$\begin{aligned}\int_0^L |x-y|dy &= \int_0^x (x-y)dy + \int_x^L (y-x)dy \\ &= \frac{x^2}{2} + \frac{L^2}{2} - \frac{x^2}{2} - x(L-x) \\ &= \frac{L^2}{2} + x^2 - xL.\end{aligned}$$

Therefore,

$$E(|X-Y|) = \frac{1}{L^2}\int_0^L \Big(\frac{L^2}{2} + x^2 - xL\Big)dx = \frac{L}{3}.$$

$\square$

Expectation of Sums and Products

Remark

If $h(X,Y) = X$*, then we have that* $E[h(X,Y)] = E[X] \equiv \mu_X$.

Property

If $X \geqslant Y$*, then* $E(X) \geqslant E(Y)$*. Thus we have* $|E(X)| \leqslant E(|X|)$.

Proof. Since $X \geqslant Y$, $P(X - Y \geqslant 0) = 1$. Thus $E(X - Y) \geqslant 0$, i.e., $E(X) \geqslant E(Y)$.

Since $-|X| \leqslant X \leqslant |X|$, $-E(|X|) \leqslant E(X) \leqslant E(|X|)$, i.e., $|E(X)| \leqslant E(|X|)$. $\square$

Expectation of Sums and Products

Among various functions of two random variables, we mainly concern sums and products of two random variables.

Theorem

Suppose that $E(X)$ and $E(Y)$ are both finite. We have

$$E(X+Y) = E(X) + E(Y). \tag{26}$$

Proof. In the continuous case,

$$\begin{aligned} E(X+Y) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x+y)f(x,y)dxdy \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}xf(x,y)dydx + \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}yf(x,y)dxdy \\ &= \int_{-\infty}^{\infty}xf_X(x)dx + \int_{-\infty}^{\infty}yf_Y(y)dy \\ &= E(X) + E(Y). \end{aligned}$$

The proof is similar in discrete case. □

Expectation of Sums and Products

Generally, we easily prove that

$$E(aX + bY + c) = aE(X) + bE(Y) + c,$$

(if the expected values exist), where a, b, and c are real constants. For example, $Y = X^2$, then the above formula enables us to write that

$$E(aX + bX^2) = aE(X) + bE(X^2).$$

Expectation of Sums and Products

Theorem

If X and Y are two independent random variables, then

$$E(XY) = E(X)E(Y).$$

And for any functions h and g,

$$E[g(X)h(Y)] = E[g(X)]E[h(Y)].$$

Expectation of Sums and Products

Proof. Suppose that X and Y are jointly continuous with joint density $f(x, y)$, and the marginal p.d.f. of X and Y are $f_X(x)$ and $f_Y(y)$ respectively. Since X and Y are independent,

$$\begin{aligned} E[g(X)h(Y)] &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x)h(y)f(x,y)dxdy \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x)h(y)f_X(x)f_Y(y)dxdy \\ &= \int_{-\infty}^{\infty} g(x)f_X(x)dx \int_{-\infty}^{\infty} h(y)f_Y(y)dy \\ &= E[g(X)]E[h(Y)]. \end{aligned}$$

The proof is similar in the discrete case. $\square$

Expectation of Sums and Products

Recall that if X and Y are any two random variables then $E(X+Y) = E(X) + E(Y)$. This is not always true for the case of the variance.

For example, let X be a random variable with $Var(X) \neq 0$, and define $Y = -X$. Then $Var(X) = Var(Y)$, so that $Var(X) + Var(Y) = 2Var(X)$. But $X + Y$ is always 0 and hence has variance 0. Thus $Var(X+Y) \neq Var(X) + Var(Y)$. In fact, we have

Expectation of Sums and Products

Theorem

Let X and Y be two independent random variables. Then

$$Var(X+Y) = Var(X) + Var(Y).$$

Proof. According to the definition of the variance of random variables X and Y,

$$\begin{aligned}
Var(X+Y) &= E\big\{[(X+Y) - E(X+Y)]^2\big\} \\
&= E\big\{[(X - E(X)) + (Y - E(Y))]^2\big\} \\
&= E[(X - E(X))^2] + E[(Y - E(Y))^2] + 2E[(X - E(X))(\\
&= Var(X) + Var(Y) + 2[E(XY) - E(X)E(Y)].
\end{aligned}$$

Since X and Y are independent, $E(XY) = E(X)E(Y)$. Thus

$$Var(X+Y) = Var(X) + Var(Y).$$

$\square$

Expectation of Sums and Products

Corralary

If $X \perp Y$, *then*

$$Var(aX + bY + c) = a^2 Var(X) + b^2 Var(Y).$$

Note that the constant c does not influence the variance.

Expectation of Sums and Products

Example

Suppose that $X \sim N(0,1)$, $Y \sim N(0,1)$ and $X \perp Y$. Determine the value of $E(|X-Y|)$.

Solution. Let $Z = X - Y$. The distribution of Z must be normal. We can determine the parameters μ_Z and σ_Z^2 by

$$\mu_Z = E(Z) = E(X-Y) = E(X) - E(Y) = 0.$$
$$\sigma_Z^2 = Var(X-Y) = Var(X) + Var(Y) = 2.$$

i.e., $Z \sim N(0,2)$. The p.d.f. of Z is

$$f(z) = \frac{1}{\sqrt{2\pi}\sqrt{2}} \exp\{-z^2/4\}.$$

Then

$$E(|X-Y|) = E(|Z|) = \frac{1}{\sqrt{2\pi}\sqrt{2}} \int_{-\infty}^{\infty} |z| \exp\{-z^2/4\} dz = \frac{2}{\sqrt{\pi}}.$$

$\square$

Expectation of Sums and Products

Theorem

(***The Cauchy-Schwarz inequality***) *Let X and Y be jointly distributed random variables having finite second moments. Then*

$$[E(XY)]^2 \leqslant E(X^2)E(Y^2). \tag{27}$$

In addition, the equality holds if and only if $Y = aX + b$ for some numbers a and b with $a \neq 0$.

Proof. For all real λ,

$$0 \leqslant E[(X + \lambda Y)^2] = E(X^2) + 2\lambda E(XY) + \lambda^2 E(Y^2).$$

Considered as a quadratic function of λ, there is, then, at most one real root. Equivalently, the discriminant of the quadratic expression is non-positive. That is

$$4[E(XY)]^2 \leqslant 4E(X^2)E(Y^2).$$

$\square$

Section 3.6 Numerical Characteristics of Random Vectors

- 3.6.1 Expectation of Sums and Products
- 3.6.2 Covariance and Correlation

Covariance and Correlation

When two random variables X and Y are not independent, it is frequently of interest to assess how strongly they are related to one another. In section 3.6.1, we have

$$Var(X+Y) = Var(X) + Var(Y) + 2E[(X - E(X))(Y - E(Y))].$$

Now we define the covariance of X and Y by the cross-product term. Let $\mu_X = E(X)$ and $\mu_Y = E(Y)$.

Covariance and Correlation

Definition

The ***covariance*** *between two random variables X and Y is*

$$Cov(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$
$$= \begin{cases} \sum_x \sum_y (x-\mu_X)(y-\mu_Y)\cdot p(x,y), & X,\ Y \text{ discrete}, \\ \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (x-\mu_X)(y-\mu_Y)\cdot f(x,y)dxdy, & X,\ Y \text{ continuous}. \end{cases}$$

Remark: This is the most important in the central joint moments, which are

$$\sigma_{ij} = E[(X-E(X))^i(Y-E(Y))^j], \quad i,j \geqslant 1.$$

Covariance and Correlation

In fact, we usually use the following way to calculate the value of covariance.

$$\begin{aligned} Cov(X,Y) &= E[XY - \mu_X Y - X\mu_Y + \mu_X\mu_Y] \\ &= E(XY) - \mu_X\mu_Y - \mu_X\mu_Y + \mu_X\mu_Y \\ &= E(XY) - \mu_X\mu_Y. \end{aligned}$$

Covariance and Correlation

The rationale for the definition is as follows.

Suppose that X and Y have a strong positive relationship to one another, by which we mean that large values of X tend to occur with large values of Y and small values of X with small values of Y. Then most of the probability mass or density will be associated with $(x - \mu_X)$ and $(y - \mu_Y)$, either both positive (both X and Y above their respective means) or both negative, so the product $(x - \mu_X)(y - \mu_Y)$ will tend to be positive.

Covariance and Correlation

Thus for a strong positive relationship, $Cov(X, Y)$ should be quite positive. For a strong negative relationship, the signs of $(x - \mu_X)$ and $(y - \mu_Y)$ will tend to be opposite, yielding a negative product. Thus for a strong negative relationship, $Cov(X, Y)$ should be quite negative. If X and Y are not strongly related, positive and negative products will tend to cancel one another, yielding a covariance near 0.

Covariance and Correlation

Let us consider now the special case where X and Y are indicator variables for whether or not the events A and B occur. That is, for events A and B, define

$$X = \begin{cases} 1 & \text{if } A \text{ occurs,} \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad Y = \begin{cases} 1 & \text{if } B \text{ occurs,} \\ 0 & \text{otherwise.} \end{cases}$$

Because XY will equal 1 or 0 depending on whether or not both X and Y equal 1, we see that

$$Cov(X, Y) = P(X = 1, Y = 1) - P(X = 1)P(Y = 1).$$

From this it follows that

Covariance and Correlation

$$
\begin{aligned}
Cov(X,Y) > 0 &\iff P(X=1,Y=1) > P(X=1)P(Y=1) \\
&\iff \frac{P(X=1,Y=1)}{P(X=1)} > P(Y=1) \\
&\iff P(Y=1|X=1) > P(Y=1).
\end{aligned}
$$

That is, the covariance of X and Y is positive if the outcome $X = 1$ makes it more likely that $Y = 1$ (which, as is easily seen by symmetry, also implies the reverse). In general it can be shown that a positive value of $Cov(X,Y)$ is an indication that Y tends to increase as X does, whereas a negative value indicates that Y tends to decrease as X increases.

Covariance and Correlation

Theorem

For jointly distributed random variables X and Y, and constants a, b, c, d, we have
(i) $Cov(X, Y) = Cov(Y, X)$,
(ii) $Cov(X, X) = E(X^2) - [E(X)]^2 = Var(X)$,
(iii) $Cov(aX + b, cY + d) = acCov(X, Y)$,
(iv) $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$.

Proof. Parts (i), (ii) and (iv) are obvious.

$$\begin{aligned} & Cov(aX + b, cY + d) \\ = & E[(aX + b - aE(X) - b)(cY + d - cE(Y) - d)] \\ = & E[ac(X - E(X))(Y - E(Y))] = acCov(X, Y). \end{aligned}$$

Parts (iii) is proved as required. □

Covariance and Correlation

Example

Consider the joint p.d.f. of (X, Y) is

$$f(x,y) = \begin{cases} 24xy & \text{for } 0 \leqslant x \leqslant 1, 0 \leqslant y \leqslant 1, x+y \leqslant 1, \\ 0 & \text{otherwise.} \end{cases}$$

Calculate the value of $Cov(X, Y)$.

Solution. From the joint p.d.f. $f(x, y)$ of (X, Y), the marginal distribution of X is

$$f_X(x) = \begin{cases} 12x(1-x)^2 & \text{for } 0 \leqslant x \leqslant 1, \\ 0 & \text{otherwise.} \end{cases}$$

And $f_Y(y)$ can be obtained by replacing x by y in $f_X(x)$.

Covariance and Correlation

It is easily verified that $\mu_X = \mu_Y = 2/5$, and

$$\begin{aligned} E(XY) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xyf(x,y)dxdy \\ &= \int_0^1\int_0^{1-x} xy \cdot 24xydydx \\ &= 8\int_0^1 x^2(1-x)^3dx = \frac{2}{15}. \end{aligned}$$

Thus $Cov(X,Y) = E(XY) - E(X)E(Y) = 2/15 - (2/5)(2/5) = -2/75$. $\square$

Covariance and Correlation

Example

$x \backslash y$	0	100	200
100	0.2	0.1	0.2
250	0.05	0.15	0.3

x	100	250
$p_X(x)$	0.5	0.5

y	0	100	200
$p_Y(y)$	0.25	0.25	0.5

Calculate the value of $Cov(X, Y)$.

Covariance and Correlation

Solution. It can be easily calculated that $\mu_X = \sum x p_X(x) = 175$ and $\mu_Y = 125$. Therefore,

$$
\begin{aligned}
& Cov(X, Y) \\
= & \sum_{\text{all } (x,y)} (x - 175)(y - 125) p(x, y) \\
= & (100 - 175)(0 - 125) \cdot 0.2 + \cdots + (250 - 175)(200 - 125) \cdot 0.3 \\
= & 1875.
\end{aligned}
$$

$\square$

Covariance and Correlation

It might appear that the relationship in the insurance example is quite strong since $Cov(X, Y) = 1875$, whereas $Cov(X, Y) = -2/75$ in the nut example would seem to imply quite a weak relationship. Unfortunately, the covariance has a serious defect that makes it impossible to interpret a computed value. In the insurance example, suppose we had expressed the deductible amount in cents rather than in dollars. Then $100X$ would replace X, $100Y$ would replace Y, and the resulting covariance would be

$$Cov(100X, 100Y) = (100)(100)Cov(X, Y) = 18,750,000.$$

If, on the other hand, the deductible amount had been expressed in hundreds of dollars, the computed covariance would have been $(0.01)(0.01)(1875)$ =0.1875.

Covariance and Correlation

Although the sign of the covariance tells us whether two random variables have a positive or negative relationship, its magnitude and units are difficult to interpret. To avoid this problem, we introduce the definition of correlation coefficient.

Definition

The ***correlation coefficient*** *of X and Y, denoted by $\rho(X,Y)$, ρ_{XY}, or just ρ, is defined by*

$$\rho(X,Y) = \rho_{XY} = \frac{Cov(X,Y)}{\sigma_X \cdot \sigma_Y},$$

where σ_X and σ_Y are standard deviation of X and Y respectively.

Covariance and Correlation

Example

Suppose that $E(X) = -2$, $E(Y) = 4$, $Var(X) = 4$ and $Var(Y) = 9$. Let $Z = 3X^2 - 2XY + Y^2 - 3$. Find the value of $E(Z)$ according to following conditions: (a) $X \perp Y$ and (b) $\rho_{XY} = 0.5$.

Solution. (a) Since $X \perp Y$, $E(XY) = E(X)E(Y)$. Thus

$$\begin{aligned} E(Z) &= E(3X^2 - 2XY + Y^2 - 3) \\ &= 3E(X^2) - 2E(X)E(Y) + E(Y^2) - 3 \\ &= 3[Var(X) + [E(X)]^2] - 2E(X)E(Y) + \\ &\quad [Var(Y) + [E(Y)]^2] - 3 = 62. \end{aligned}$$

Covariance and Correlation

(b) Using $Cov(X,Y) = \rho(X,Y)\sqrt{Var(X)}\sqrt{Var(Y)}$, we have

$$\begin{aligned} E(Z) &= 3[Var(X) + [E(X)]^2] - 2E(X)E(Y) + \\ &\quad [Var(Y) + [E(Y)]^2] - 3 \\ &= 24 - 2[Cov(X,Y) + E(X)E(Y)] + 25 - 3 \\ &= 24 - 2[\rho(X,Y)\sqrt{Var(X)}\sqrt{Var(Y)} + \\ &\quad E(X)E(Y)] + 25 - 3 = 68. \end{aligned}$$

$\square$

Covariance and Correlation

Property

(i) *If a and c are either both positive or both negative, then*

$$\rho(aX + b,\ cY + d) = \rho(X, Y)$$

(ii) *For any two random variables X and Y, $-1 \leqslant \rho(X, Y) \leqslant 1$. And $\rho(X, Y) = 1$ or -1 if and only if $Y = aX + b$ for some numbers a and b with $a \neq 0$.*

Proof. (i) Let σ_{aX+b} denote the variance of $aX + b$ and σ_{cY+d} the variance of $cY + d$.

$$Corr(aX + b,\ cY + d) = \frac{Cov(aX + b,\ cY + d)}{\sigma_{aX+b} \cdot \sigma_{cY+d}}$$

$$= \frac{ac\ Cov(X, Y)}{ac\ \sigma_X \sigma_Y} = Corr(X,\ Y).$$

Covariance and Correlation

(ii) By using Cauchy-Schwarz inequality,

$$|E(XY)| \leqslant \sqrt{E(X^2)}\sqrt{E(Y^2)},$$

we have

$$\begin{aligned} &\big|E\big[(X - E(X))(Y - E(Y))\big]\big| \\ &\leqslant \sqrt{E[(X - E(X)^2)]}\sqrt{E[(Y - E(Y)^2)]}. \end{aligned}$$

i.e., $|\rho(X, Y)| \leqslant \sigma_X \sigma_Y$. Thus $|\rho(X, Y)| \leqslant 1$. $\square$

Covariance and Correlation

This Proposition says that ρ is a measure of the degree of ***linear relationship*** between X and Y, and only when the two variables are perfectly related in a linear manner will ρ be as positive or negative as it can be.

ρ less than 1 in absolute value indicates only that the relationship is not completely linear, but there may still be a very strong nonlinear relation.

Also, $\rho = 0$ does not imply that X and Y are independent, but only that there is complete absence of a linear relationship. When $\rho = 0$, X and Y are said to be ***uncorrelated***.

Two variables could be uncorrelated yet highly dependent because there is a strong nonlinear relationship, so be careful not to conclude too much from knowing that $\rho = 0$. Let us see an example.

Covariance and Correlation

Example

Suppose that $X \sim U(-1/2, 1/2)$. Let $Y = \cos X$. Find the value of $\rho(X, Y)$.

Solution. It is obvious that $E(X) = 0$.

$$\begin{aligned} Cov(X, Y) &= E(XY) - E(X)E(Y) \\ &= E(X \cos X) = \int_{-1/2}^{1/2} x \cos x dx = 0. \end{aligned}$$

Then $\rho(X, Y) = 0$. $\square$

Random variables X and Y in above example are said to be uncorrelated since $\rho(X, Y) = 0$. But it is obvious that X and Y have a strong relationship.

Covariance and Correlation

Example

The joint density function of the continuous random vector (X, Y) is

$$f(x, y) = \begin{cases} e^{-2y} & \text{if } 0 < x < 2,\ y > 0, \\ 0 & \text{elsewhere.} \end{cases}$$

What is the correlation coefficient of X and Y?

Solution. X and Y are independent since

$$e^{-2y} = g(x)h(y),$$

where $g(x) = 1$ and $h(y) = e^{-2y}$. It follows from Theorem 3.6.4 that $\rho(X, Y) = 0$. $\square$

Covariance and Correlation

Example

Suppose that $(X, Y) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$. Find the correlation coefficient of X and Y.

Solution. The joint p.d.f. of (X, Y) is

$$f(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2}\right]}.$$

We have known that

$$X \sim N(\mu_1, \sigma_1^2), \quad E(X) = \mu_1, \quad Var(X) = \sigma_1^2,$$

$$Y \sim N(\mu_2, \sigma_2^2), \quad E(Y) = \mu_2, \quad Var(Y) = \sigma_2^2.$$

Covariance and Correlation

Then

$$
\begin{aligned}
Cov(X,Y) &= E[(X-\mu_1)(Y-\mu_2)] \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x-\mu_1)(y-\mu_2)f(x,y)dxdy \\
&= \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(x-\mu_1)(y-\mu_2)\exp\Big[\frac{-(y-\mu_2)^2}{2\sigma_2^2}\Big]\cdot \\
&\qquad \exp\Big\{-\frac{1}{2(1-\rho^2)}\Big[\frac{x-\mu_1}{\sigma_1}-\rho\frac{y-\mu_2}{\sigma_2}\Big]^2\Big\}dxdy.
\end{aligned}
$$

Let $t = -\frac{1}{2(1-\rho^2)}\left[\frac{x-\mu_1}{\sigma_1}-\rho\frac{y-\mu_2}{\sigma_2}\right]$ and $u = \frac{y-\mu_2}{\sigma_2}$. Then $\left|\frac{\partial(x,y)}{\partial(t,u)}\right| = \sigma_1\sigma_2\sqrt{1-\rho^2}$. Thus by equation

$$
\iint_{D_{xy}} f(x,y)dxdy = \iint_{D_{tu}} f\big(x(t,u),y(t,u)\big)\Big|\frac{\partial(x,y)}{\partial(t,u)}\Big|dtdu,
$$

Covariance and Correlation

we get

$$
\begin{aligned}
Cov(X,Y) &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(\sigma_1\sigma_2\sqrt{1-\rho^2}tu+\rho\sigma_1\sigma_2u^2)\exp\Big(-\frac{u^2}{2}-\frac{t^2}{2}\Big)dtdu \\
&= \frac{\rho\sigma_1\sigma_2}{2\pi}\int_{-\infty}^{\infty}u^2\exp\Big(-\frac{u^2}{2}\Big)du\int_{-\infty}^{\infty}\exp\Big(-\frac{t^2}{2}\Big)dt \\
&\qquad + \frac{\sigma_1\sigma_2}{2\pi}\sqrt{1-\rho^2}\int_{-\infty}^{\infty}u\exp\Big(-\frac{u^2}{2}\Big)du\int_{-\infty}^{\infty}t\exp\Big(-\frac{t^2}{2}\Big)dt \\
&= \frac{\rho\sigma_1\sigma_2}{2\pi}\sqrt{2\pi}\sqrt{2\pi} = \rho\sigma_1\sigma_2,
\end{aligned}
$$

which means $\rho(X,Y)=\rho$. $\square$

Covariance and Correlation

Now we make clear that the last parameter ρ is just the correlation coefficient of X and Y in bivariate normal distribution.

Uncorrelated and independent are usually not equivalent to each other. But normal distribution is the special case.

Property

If $(X, Y) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$, *then* $X \perp Y \Leftrightarrow \rho(X, Y) = 0$.

Covariance and Correlation

In the end of this section, let us review the expected value and variance of one random variable and contrast them to those of a random vector. Let X and Y be two random variables defined on the same sample space Ω. Then (X, Y) is a random vector on $\Omega \times \Omega$. For random variable X, we have

$$\mu = E(X), \qquad \sigma^2 = Var(X).$$

For random vector (X, Y), we have five numerical values:

$$\mu_1 = E(X),\ \mu_2 = E(Y);$$

$$\sigma_1^2 = Var(X),\ \sigma_2^2 = Var(Y);\ Cov(X, Y).$$

Covariance and Correlation

That means for a two dimensional random vector, we need consider five numerical value. Now we describe these information in matrix form. Let $\boldsymbol{X} = (X_1, X_2)'$. Then we denote

$$\mu = E(\boldsymbol{X}) = \begin{pmatrix} E(X_1) \\ E(X_2) \end{pmatrix}, \quad \Sigma = \begin{pmatrix} Var(X_1) & Cov(X_1, X_2) \\ Cov(X_2, X_1) & Var(X_2) \end{pmatrix}.$$

It can be seen that μ, the mean vector, describes the information of 1-th moments of random vector $\boldsymbol{X}$.

Σ, the covariance matrix, describes the information of 2-th central moments of random vector $\boldsymbol{X}$.

Covariance and Correlation

The expected value vector and covariance matrix of random vector (X_1, X_2) is the generalization of the mean and variance of a random variable on one hand, and they can make some probability distribution functions easier to describe on the other hand. Let us see the benefits by the example of two dimensional normal distribution.

Covariance and Correlation

Suppose that $(X_1, X_2) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$. We have known that

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2}\right]}$$

where $-\infty < x_1, x_2 < \infty$. It is indeed difficult to remember.

By using

$$\boldsymbol{x} = (x_1, x_2)', \ \mu = (\mu_1, \mu_2)', \ \Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix},$$

we can transform the above p.d.f. to the following form:

$$f(x, y) = \frac{1}{(2\pi)^{2/2}|\Sigma|^{1/2}} \exp\{-\frac{1}{2}(\boldsymbol{x} - \mu)'\Sigma^{-1}(\boldsymbol{x} - \mu)\},$$

Covariance and Correlation

where $|\Sigma| = \sigma_1^2\sigma_2^2(1-\rho^2)$ is the determinant of Σ, and

$$\Sigma^{-1} = \frac{1}{|\Sigma|}\begin{pmatrix} \sigma_2^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_1^2 \end{pmatrix}$$

is the inverse matrix of Σ. That means bivariate normal distribution is determined by the expected vector and covariance matrix. We can denote $\boldsymbol{X} \sim N(\mu, \Sigma)$.

It is easy to memorize the p.d.f. than before, isn't it? It can help us to generalize two dimensional normal distribution to multiple normal distribution.

Section 3.7 Multivariate Distributions

Suppose that we have n random variables. When working with all these random variables together, we will often find it convenient to put them in a vector $\boldsymbol{X} = (X_1, X_2, \cdots, X_n)'$. This random vector is a mapping from Ω^n to $\mathbb{R}^n$.
If $X_1, X_2, \cdots, X_n$ are all discrete random variables, then the joint p.f. of the random vector $\boldsymbol{X}$ is the function

$$p(x_1, x_2, \cdots, x_n) = P(X_1 = x_1, X_2 = x_2, \cdots, X_n = x_n).$$

If the variables are continuous, then the joint p.d.f. of $\boldsymbol{X}$ is the function $f(x_1, x_2, \cdots, x_n)$ such that for any n intervals $[a_1, b_1]$, $\cdots$, $[a_n, b_n]$,

$$P(a_1 \leqslant X_1 \leqslant b_1, \cdots, a_n \leqslant X_n \leqslant b_n)$$

$$= \int_{a_1}^{b_1} \cdots \int_{a_n}^{b_n} f(x_1, x_2, \cdots, x_n) dx_1 \cdots dx_2.$$

Section 3.7 Multivariate Distributions

- 3.7.1 Distribution Functions of Multiple Random Vectors
- 3.7.2 Numerical Characteristics of Random Vectors
- 3.7.3 Multiple Normal Distribution

Distribution Functions of Multiple Random Vectors

Since the discrete cases are relatively straightforward, here we mainly introduce the continuous cases.

It should be clear that random vectors are simply an alternative notation for dealing with n random variables, so the notions of joint p.d.f. and d.f. will apply to random vectors as well. we can define the joint distribution function, the joint probability density function , the marginal probability density function of X_i, and the conditional probability density function of X_i given $X_1, \cdots, X_{i-1}, X_{i+1}, \cdots, X_n$, respectively, as

Distribution Functions of Multiple Random Vectors

$$F(x_1, x_2, \cdots, x_n) = P(X_1 \leqslant x_1, X_2 \leqslant x_2, \cdots, X_n \leqslant x_n),$$

$$f(x_1, x_2, \cdots, x_n) = \frac{\partial^n F(x_1, x_2, \cdots, x_n)}{\partial x_1 \cdots \partial x_n},$$

$$f_{X_i}(x_i) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, \cdots, x_n) dx_1 \cdots dx_{i-1} dx_{i+1} \cdots dx_n,$$

$$f_{X_i|X_1,\cdots,X_{i-1},X_{i+1},\cdots,X_n}(x_i|x_1, \cdots, x_{i-1}, x_{i+1}, \cdots, x_n)$$

$$= \frac{f(x_1, x_2, \cdots, x_n)}{f_{X_1,\cdots,X_{i-1},X_{i+1},\cdots,X_n}(x_1, \cdots, x_{i-1}, x_{i+1}, \cdots, x_n)}.$$

Distribution Functions of Multiple Random Vectors

where

$$f_{X_1,\cdots,X_{i-1},X_{i+1},\cdots,X_n}(x_1,\cdots,x_{i-1},x_{i+1},\cdots,x_n)$$

$$= \int_{-\infty}^{+\infty} f(x_1,\cdots,x_n)dx_i.$$

To calculate the probability of an event $A \subseteq R^n$ we have,

$$P((x_1,x_2,\cdots,x_n) \in A)$$

$$= \int \cdots \int_{(x_1,x_2,\cdots,x_n)\in A} f(x_1,x_2,\cdots,x_n)dx_1dx_2\cdots dx_n.$$

Distribution Functions of Multiple Random Vectors

Remark

In n dimensions, a continuous random vector possesses a joint density function defined on $\mathbb{R}^n$ (or on an uncountably infinite subset of $\mathbb{R}^n$). This function is nonnegative and its integral on $\mathbb{R}^n$ is equal to 1.

Definition

The random variables $X_1, X_2, \cdots, X_n$ are said to be independent if for every subset $X_{i1}, X_{i2}, \cdots, X_{ik}$ of the variables (each pair, each triple, and so on), the joint p.f. or p.d.f. of the subset is equal to the product of the marginal p.f.'s or p.d.f.'s.

Distribution Functions of Multiple Random Vectors

Thus if the variables are independent with $n = 4$, then the joint p.f. or p.d.f. of any two variables is the product of the two marginals, and similarly for any three variables and all four variables together. Most important, once we are told that n variables are independent, then the joint p.f. or p.d.f. is the product of the n marginals.

Distribution Functions of Multiple Random Vectors

Property

(i) *If* $X_1,\ X_2, \cdots,\ X_m$ *are independent and* X_i *has a Bernoulli distribution with parameter* p*, for* $i = 1, 2,\ \cdots, n$*, then we have:*

$$\sum_{i=1}^{n} X_i \sim B(n,\ p), \quad for\ n = 1, 2, \cdots$$

More generally, if $X_1,\ X_2, \cdots,\ X_m$ *are independent, and* $X_i \sim B(n_i, p)$*,* $i = 1, 2, \cdots, m$*. Then* $X_1 + X_2 + \cdots + X_m \sim B(n_1 + n_2 + \cdots + n_m,\ p)$*.*

(ii) *Assume* $X_1,\ X_2, \cdots,\ X_n$ *are independent, and* $X_i \sim P(\lambda_i)$*,* $i = 1, 2, \cdots, n$*. Then* $X_1 + X_2 + \cdots + X_n \sim P(\lambda_1 + \cdots + \lambda_n)$*.*

Distribution Functions of Multiple Random Vectors

Property

(iii) *Assume* $X_1, X_2, \cdots, X_n$ *are independent, and* $X_i \sim Exp(\lambda)$, *for* $i = 1, \cdots, n$, *then*

$$\sum_{i=1}^{n} a_i X_i \sim G(n, \lambda).$$

(iv) *Assume* $X_1, X_2, \cdots, X_n$ *are independent, and* $X_i \sim N(\mu_i, \sigma_i^2)$, *for* $i = 1, \cdots, n$, *then*

$$\sum_{i=1}^{n} a_i X_i \sim N\Big(\sum_{i=1}^{n} a_i \mu_i, \sum_{i=1}^{n} a_i^2 \sigma_i^2\Big).$$

Distribution Functions of Multiple Random Vectors

Example

The total revenue from the sale of the three grades of gasoline on a particular day was $Y = 21.2X_1 + 21.35X_2 + 21.5X_3$, and we calculated $\mu_Y = 4125$ and $\sigma_Y = 322.53$. If the X_i $i = 1, 2, 3$ are normally distributed (assuming independence), what is the probability that revenue exceeds 4500?

Solution. Since Y is the linear combination of three independent random variables, the distribution of Y must be normal distribution. Then

$$\begin{aligned} P(Y > 4500) &= P\Big(\frac{Y - 4125}{322.53} > \frac{4500 - 4125}{322.53}\Big) \\ &= 1 - \Phi(1.16) = 0.1230. \end{aligned}$$

$\square$

Distribution Functions of Multiple Random Vectors

Next, suppose that $X_1, X_2, \cdots, X_n$ are independent and the distribution function of X_i is F_{X_i} $(i = 1, 2, \cdots, n)$. We consider the distributions of $Z = \max\{X_1, X_2, \cdots X_n\}$ and $W = \min\{X_1, X_2, \cdots X_n\}$. Then

$$F_Z(z) = F_{X_1}(z)F_{X_2}(z)\cdots F_{X_n}(z), \tag{28}$$

$$F_W(w) = 1 - [1 - F_{X_1}(w)][1 - F_{X_2}(w)]\cdots[1 - F_{X_n}(w)]. \tag{29}$$

Distribution Functions of Multiple Random Vectors

Especially, if $X_1, X_2, \cdots, X_n$ are independent and have identical distribution function F and probability density function f, then

$$F_Z(z) = [F(z)]^n, \tag{30}$$

$$\text{and} \quad F_W(w) = 1 - [1 - F(w)]^n. \tag{31}$$

Hence,

$$f_Z(z) = n[F(z)]^{n-1} f(z), \tag{32}$$

$$\text{and} \quad f_W(w) = n[1 - F(w)]^{n-1} f(w). \tag{33}$$

Distribution Functions of Multiple Random Vectors

Example

Suppose that $X_1, \cdots, X_n$ represent the lifetimes of n components, the components operate independently of one another, and each lifetime is exponentially distributed with parameter λ.
(i) if these n components constitute a system that will fail as soon as a single component fails, then calculate the probability that the system lasts past time t.
(ii) if these n components constitute a system that will fail when the last single component fails, then calculate the probability that the system lasts past time t.

Distribution Functions of Multiple Random Vectors

Solution. (i) We calculate the probability by two methods. First, the distribution of X_i is

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geqslant 0, \\ 0 & \text{otherwise.} \end{cases}$$

Since $X_1, X_2, \cdots, X_n$ are independent,

$$\begin{aligned} f(x_1, x_2, \cdots, x_n) &= (\lambda e^{-\lambda x_1}) \cdot (\lambda e^{-\lambda x_2}) \cdot \cdots \cdot (-\lambda e^{\lambda x_n}) \\ &= \begin{cases} \lambda^n e^{-\lambda \sum x_i} & \text{for } x_1 \geqslant 0,\ x_2 \geqslant 0\ \cdots,\ x_n \geqslant 0, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

For system (i), the probability that the system lasts past time t is

$$\begin{aligned} P(X_1 > t, \cdots, X_n > t) &= \int_t^\infty \cdots \int_t^\infty f(x_1, \cdots, x_n) dx_1 \cdots dx_n \\ &= \Big(\int_t^\infty \lambda e^{-\lambda x_1} dx_1 \Big) \cdots \Big(\int_t^\infty \lambda e^{-\lambda x_n} dx_n \Big) \\ &= (e^{-\lambda t})^n = e^{-n\lambda t}. \end{aligned}$$

Distribution Functions of Multiple Random Vectors

Alternatively, by using equation (31),

$$\begin{aligned} P(\text{system lifetime} \leqslant t) =& P(\min\{X_1, X_2, \cdots, X_n\} \leqslant t) \\ =& 1 - e^{-n\lambda t}, \quad \text{for } t \geqslant 0. \end{aligned}$$

Then the probability that the system lasts past time t is $1 - P(\text{system lifetime} \leqslant t) = e^{-n\lambda t}$. which shows that system lifetime has an exponential distribution with parameter $n\lambda$; the expected value of system lifetime is $1/n\lambda$.

Distribution Functions of Multiple Random Vectors

(ii) For system (ii), by using equation (30),

$$\begin{aligned} P(\text{system lifetime} \leqslant t) =& P(\max\{X_1, X_2, \cdots, X_n\} \leqslant t) \\ =& (1 - e^{-\lambda t})^n, \quad \text{for } t \geqslant 0. \end{aligned}$$

Then the probability that the system lasts past time t is

$$1 - P(\text{system lifetime} \leqslant t) = 1 - (1 - e^{-\lambda t})^n.$$

$\square$

Section 3.7 Multivariate Distributions

- 3.7.1 Distribution Functions of Multiple Random Vectors
- 3.7.2 Numerical Characteristics of Random Vectors
- 3.7.3 Multiple Normal Distribution

Numerical Characteristics of Random Vectors

Suppose that the random vector $\boldsymbol{X} = (X_1, X_2, \cdots, X_n)'$ has the distribution $f(x_1, x_2, \cdots, x_n)$. Consider an arbitrary function from $g : \mathbb{R}^n \to \mathbb{R}$. The expected value of this function is defined as

$$E[g(\boldsymbol{X})] = \int_{\mathbb{R}^n} g(x_1, x_2, \cdots, x_n) f(x_1, x_2, \cdots, x_n) dx_1 dx_2 \cdots dx_n, \tag{34}$$

where $\int_{\mathbb{R}^n}$ is n consecutive integrations from $-\infty$ to ∞.

Numerical Characteristics of Random Vectors

The expectation $\mu = E(\boldsymbol{X})$ and covariance matrix Σ of random vector $\boldsymbol{X}$ are defined by

$$E(\boldsymbol{X}) = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{pmatrix} = \begin{pmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_n) \end{pmatrix}$$

Numerical Characteristics of Random Vectors

and

$$\Sigma = \begin{pmatrix} Var(X_1^2) & \cdots & Cov(X_1, X_n) \\ Cov(X_2, X_1) & \cdots & Cov(X_2, X_n) \\ \vdots & \ddots & \vdots \\ Cov(X_n, X_1) & \cdots & Var(X_n^2) \end{pmatrix}, \tag{35}$$

Numerical Characteristics of Random Vectors

where

$$\mu_i = E(X_i) = \int_{\mathbb{R}^n} x_i f(x_1, x_2, \cdots, x_n) dx_1 dx_2 \cdots dx_n,$$

$$Cov(X_i, X_j) = \int_{\mathbb{R}^n} (x_i - \mu_i)(x_i - \mu_j) f(x_1, x_2, \cdots, x_n) dx_1 dx_2 \cdots dx_n,$$

$i, j = 1, 2, \cdots$.

Generally, if g is a function from $\mathbb{R}^n$ to $\mathbb{R}^m$, then the expected value of g is the element-wise expected values of the output vector. i.e., if g is $g(\boldsymbol{x}) = (g_1(\boldsymbol{X}), g_2(\boldsymbol{X}), \cdots, g_m(\boldsymbol{X}))'$, then $E[g(\boldsymbol{X})] = (E[g_1(\boldsymbol{X})], E[g_2(\boldsymbol{X})], \cdots, E[g_m(\boldsymbol{X})])'$.

Numerical Characteristics of Random Vectors

Property

$\Sigma = E[(\boldsymbol{X} - E(\boldsymbol{X}))(\boldsymbol{X} - E(\boldsymbol{X}))']$.

From the Proposition, we can see the covariance matrix is the generalization of variance.

Numerical Characteristics of Random Vectors

Suppose that $X_1, X_2, \cdots, X_n$ are independent. To obtain the distribution of $X_1 + X_2 + \cdots + X_n$, we need calculate the distribution of the convolution n times. Although we obtain some distributions of special cases, it is really difficult to deal with the convolution in general. Often we can know the mean values of the sum and the production by using the follow proposition.

Numerical Characteristics of Random Vectors

Property

(i) *For any n random variables $X_1, X_2 \cdots, X_n$,*

$$E(X_1 + X_2 + \cdots + X_n) = E(X_1) + E(X_2) + \cdots + E(X_n). \quad (36)$$

Generally,

$$E(a_1X_1 + a_2X_2 + \cdots + a_nX_n + b)$$
$$= a_1E(X_1) + a_2E(X_2) + \cdots + a_nE(X_n) + b.$$

Moreover, for $g_i : \mathbb{R}^2 \to \mathbb{R}$, $i = 1, 2, \cdots, n$,

$$E[a_1g_1(X,Y) + a_2g_2(X,Y) + \cdots + a_ng_n(X,Y)]$$
$$= a_1E[g_1(X,Y)] + a_2E[g_2(X,Y)] + \cdots + a_nE[g_n(X,Y)].$$

where $a_i \in \mathbb{R}$ are constants, $i = 1, 2, \cdots, n$.

Numerical Characteristics of Random Vectors

Property

(ii) *If* $X_1, X_2 \cdots, X_n$ *are independent, then*

$$E(X_1 X_2 \cdots X_n) = E(X_1)E(X_2)\cdots E(X_n).$$

Property

$$Cov\Big(\sum_{i=1}^{n} X_i, \sum_{j=1}^{m} Y_j\Big) = \sum_{i=1}^{n}\sum_{j=1}^{m} Cov(X_i, Y_j).$$

Numerical Characteristics of Random Vectors

Proof. Let $\mu_i = E(X_i)$ and $\nu_j = E(Y_j)$. Then

$$E\Big(\sum_{i=1}^{n} X_i\Big) = \sum_{i=1}^{n} \mu_i, \quad E\Big(\sum_{j=1}^{m} Y_j\Big) = \sum_{j=1}^{m} \nu_j$$

and

$$\begin{aligned}
Cov\Big(\sum_{i=1}^{n} X_i, \sum_{j=1}^{m} Y_j\Big) &= E\Big[\Big(\sum_{i=1}^{n} X_i - \sum_{i=1}^{n} \mu_i\Big)\Big(\sum_{j=1}^{m} Y_j - \sum_{j=1}^{m} \nu_i\Big)\Big] \\
&= E\Big[\sum_{i=1}^{n}(X_i - \mu_i)\sum_{j=1}^{m}(Y_j - \nu_i)\Big] \\
&= E\Big[\sum_{i=1}^{n}\sum_{j=1}^{m}(X_i - \mu_i)(Y_j - \nu_i)\Big] \\
&= \sum_{i=1}^{n}\sum_{j=1}^{m} E[(X_i - \mu_i)(Y_j - \nu_i)],
\end{aligned}$$

where the last equality follows because the expected value of a sum of random variables is equal to the sum of the expected values. $\square$

Numerical Characteristics of Random Vectors

Property

For any n random variables $X_1, X_2 \cdots, X_n$,

$$Var\Big(\sum_{i=1}^{n} X_i\Big) = \sum_{i=1}^{n} Var(X_i) + 2\sum_{j}\sum_{i<j} Cov(X_i, X_j). \quad (37)$$

If $X_1, \cdots, X_n$ are pairwise independent, i.e., X_i and X_j are independent for $i \neq j$, then

$$Var\Big(\sum_{i=1}^{n} X_i\Big) = \sum_{i=1}^{n} Var(X_i), \quad (38)$$

and generally,

$$Var(a_1X_1 + a_2X_2 + \cdots + a_nX_n + c)$$

$$= a_1^2 Var(X_1) + a_2^2 Var(X_2) + \cdots + a_n^2 Var(X_n).$$

Numerical Characteristics of Random Vectors

Example

Suppose that N people throw their hats into the center of a room. The hats are mixed up, and each person randomly selects one. Find the expected number of people that select their own hat.

Solution. Letting X denote the number of matches, we can compute $E(X)$ most easily by writing

$$X = X_1 + X_2 + \cdots + X_n,$$

where

$$X_i = \begin{cases} 1 & \text{if the } i\text{th person selects his own hat,} \\ 0 & \text{otherwise.} \end{cases}$$

Numerical Characteristics of Random Vectors

Since, for each i, the ith person is equally likely to select any of the N hats,

$$E(X_i) = P(X_i = 1) = \frac{1}{N}.$$

Thus,

$$E(X) = E(X_1) + \cdots + E(X_N) = \frac{1}{N} \cdot N = 1.$$

Hence, on the average, exactly one person selects his own hat. $\square$

Numerical Characteristics of Random Vectors

Example

Let X be a binomial random variable with parameters n and p. Recalling that such a random variable represents the number of successes in n independent trials when each trial has probability p of being a success, we have that $X = X_1 + X_2 + \cdots + X_n$ where

$$X_i = \begin{cases} 1 & \text{if the } i\text{th trial is a success,} \\ 0 & \text{if the } i\text{th trial is a failure.} \end{cases}$$

Hence, X_i is a Bernoulli random variable having expectation $E(X_i) = 1 \cdot p + 0 \cdot (1-p)$, $Var(X_i) = E(X_i^2) - [E(X_i)]^2 = p(1-p)$. Thus,

$$E(X) = E(X_1) + \cdots + E(X_n) = np,$$
$$Var(X) = Var(X_1) + Var(X_2) + \cdots + Var(X_n) = np(1-p).$$

$\square$

Example

If independent trials having a constant probability p of being successes are performed, then determine the expected number of trials required to amass a total of r successes.

Solution. If X denotes the number of trials needed to amass a total of r successes, then X is a negative binomial random variable that can be represented by

$$X = X_1 + X_2 + \cdots + X_r,$$

where X_1 is the number of trials required to obtain the first success, X_2 the number of additional trials until the second success is obtained, X_3 the number of additional trials until the third success is obtained, and so on.

Numerical Characteristics of Random Vectors

That is, X_i represents the number of additional trials required after the $(i-1)$ success until a total of i successes is amassed. A little thought reveals that each of the random variables X_i is a geometric random variable with parameter p. Hence, $E(X_i) = 1/p$. Thus,

$$E(X) = E(X_1) + \cdots + E(X_r) = \frac{r}{p}.$$

$\square$

Numerical Characteristics of Random Vectors

Example

Suppose that there are N different types of coupons, and each time one obtains a coupon, it is equally likely to be any one of the N types. Find the expected number of coupons one need amass before obtaining a complete set of at least one of each type.

Solution. Let X denote the number of coupons collected before a complete set is attained. Define $X_i, i = 0, 1, \cdots, N-1$ to be the number of additional coupons that need be obtained after i distinct types have been collected in order to obtain another distinct type, and we note that

$$X = X_0 + X_1 + \cdots + X_{N-1}.$$

Numerical Characteristics of Random Vectors

When i distinct types of coupons have already been collected, a new coupon obtained will be of a distinct type with probability $(N-i)/N$. Therefore,

$$P(X_i = k) = \frac{N-i}{N}\left(\frac{i}{N}\right)^{k-1}, \; k \geqslant 1,$$

or, in other words, X_i is a geometric random variable with parameter $(N-i)/N$. Hence,

$$E(X_i) = \frac{N}{N-i}$$

and therefore

$$\begin{aligned} E(X) &= 1 + \frac{N}{N-1} + \frac{N}{N-2} + \cdots + \frac{N}{1} \\ &= N\left[1 + \cdots + \frac{1}{N-1} + \frac{1}{N}\right]. \end{aligned}$$

$\square$

Numerical Characteristics of Random Vectors

Property

Let $X_1, \cdots, X_n$ be independent and identically distributed random variables having distribution function F and expected value μ. Such a sequence of random variables is said to constitute a sample from the distribution F. The quantity

$$\overline{X}_n = \sum_{i=1}^{n} \frac{X_i}{n}$$

is called the ***sample mean****. Then*
(i) $E(\overline{X}_n) = \mu$,
(ii) $Var(\overline{X}_n) = \dfrac{\sigma^2}{n}$ *and* $\sigma_{\overline{X}_n} = \sigma/\sqrt{n}$.
In addition, with $S_n = X_1 + \cdots + X_n$ *(the* ***sample total****),* $E(S_n) = n\mu, Var(S_n) = n\sigma^2$, *and* $\sigma_{S_n} = \sqrt{n}\sigma$.

Numerical Characteristics of Random Vectors

According to result $E(\overline{X}_n) = \mu$, the sampling (i.e., probability) distribution of $\overline{X}_n$ is centered precisely at the mean of the population from which the sample has been selected. Result $Var(\overline{X}_n) = \dfrac{\sigma^2}{n}$ shows that the $\overline{X}_n$ distribution becomes more concentrated about μ as the sample size n increases.

Numerical Characteristics of Random Vectors

Example

In a notched tensile fatigue test on a titanium specimen, the expected number of cycles to first acoustic emission (used to indicate crack initiation) is $\mu = 28,000$, and the standard deviation of the number of cycles is $\sigma = 5000$. Let $X_1, X_2, \cdots, X_{25}$ be a random sample of size 25, where each X_i is the number of cycles on a different randomly selected specimen. Then the expected value of the sample mean number of cycles until first emission is $E(\overline{X}_{25}) = \mu = 28,000$, and the expected total number of cycles for the 25 specimens is $E(S_n) = n\mu = 25 \cdot 28,000 = 700,000$. The standard deviations of $\overline{X}_{25}$ and S_{25} are

$$\sigma_{\overline{X}_{25}} = \sigma/\sqrt{n} = \frac{5000}{\sqrt{25}} = 1000,$$

$$\sigma_{S_{25}} = \sqrt{n}\sigma = \sqrt{25} \cdot 5000 = 25,000.$$

Numerical Characteristics of Random Vectors

If the sample size increases to $n = 100$, then $E(\overline{X}_{25})$ is unchanged, but $\sigma_{\overline{X}_{25}} = 500$, half of its previous value (the sample size must be quadrupled to halve the standard deviation of $\overline{X}_{25}$). $\square$

Section 3.7 Multivariate Distributions

- 3.7.1 Distribution Functions of Multiple Random Vectors
- 3.7.2 Numerical Characteristics of Random Vectors
- 3.7.3 Multiple Normal Distribution

Multiple Normal Distribution

Let $Z_1, \cdots, Z_n$ be a set of n independent standard normal random variables. If, for some constants a_{ij}, $1 \leqslant i \leqslant m$, $1 \leqslant j \leqslant n$, and μ_i, $1 \leqslant i \leqslant m$,

$$X_1 = a_{11}Z_1 + \cdots + a_{1n}Z_n + \mu_1$$
$$X_2 = a_{21}Z_1 + \cdots + a_{2n}Z_n + \mu_2$$
$$\vdots$$
$$X_i = a_{i1}Z_1 + \cdots + a_{in}Z_n + \mu_i$$
$$\vdots$$
$$X_m = a_{m1}Z_1 + \cdots + a_{mn}Z_n + \mu_m$$

Multiple Normal Distribution

then the random vector $\boldsymbol{X} = (X_1, \cdots, X_m)$ is said to have a ***multivariate normal distribution***. We denote it by $\boldsymbol{X} \sim N(\mu, \Sigma)$, where

$$\mu = E(\boldsymbol{X}) = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_m \end{pmatrix} = \begin{pmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_m) \end{pmatrix}$$

and

$$\Sigma = \begin{pmatrix} Var(X_1^2) & \cdots & Cov(X_1, X_m) \\ Cov(X_2, X_1) & \cdots & Cov(X_2, X_m) \\ \vdots & \ddots & \vdots \\ Cov(X_m, X_1) & \cdots & Var(X_m^2) \end{pmatrix}.$$

Multiple Normal Distribution

The joint p.d.f. of $\boldsymbol{X}$ is

$$f(x,y) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\{-\frac{1}{2}(\boldsymbol{x}-\mu)'\Sigma^{-1}(\boldsymbol{x}-\mu)\},$$

where $\boldsymbol{x} = (x_1, x_2, \cdots x_n)'$.
From the fact that the sum of independent normal random variables is itself a normal random variable, it follows that each X_i is a normal random variable with mean and variance given, respectively, by

$$E(X_i) = \mu_i, \quad Var(X_i) = \sum_j^n a_{ij}.$$

When $m = 2$, the multivariate normal distribution reduces to the bivariate normal.

Multiple Normal Distribution

Property

Suppose that $X_1, \cdots, X_n$ *are* n *random variables. Then* $(X_1, \cdots, X_n)$ *has a multivariate normal distribution if and only if each linear combination of* $X_1, \cdots, X_n$ *has a normal distribution.*

Property

If $\boldsymbol{Y} = \mathsf{A}\boldsymbol{X} + \mathsf{b}$, *where* A *is an* $n \times n$ *non-singular matrix and* b *is a (column)* n*-vector of constants, then* $\boldsymbol{Y} \sim N(\mathsf{A}\mu + \mathsf{b},\ \mathsf{A}\Sigma\mathsf{A}')$, *where* $\mu = E(\boldsymbol{X})$ *and* Σ *is the covariance matrix of the multivariate normal random variable* $\boldsymbol{X}$.

Multiple Normal Distribution

Property

Suppose that $(X_1, \cdots, X_n)$ *has a multivariate normal distribution. Then* $X_1, \cdots, X_n$ *are independent are if and only if* $X_1, \cdots, X_n$ *are pairwise uncorrelated.*

Multiple Normal Distribution

Example

Suppose that $(X, Y) \sim N(0, 1, 9, 16, -1/2)$. Let $Z = \frac{X}{3} + \frac{Y}{2}$. Are X and Z independent?

Solution. (X, Z) has a bivariate normal distribution since (X, Y) has a bivariate normal distribution and

$$\begin{pmatrix} X \\ Z \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1/3 & 1/2 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}.$$

From the distribution of (X, Y), we know $Var(X) = 9$, $Var(Y) = 16$ and $\rho(X, Y) = -1/2$. Then

Multiple Normal Distribution

$$
\begin{aligned}
Cov(X, Z) &= Cov(X, \frac{X}{3} + \frac{Y}{2}) \\
&= \frac{1}{3} Cov(X, X) + \frac{1}{2} Cov(X, Y) \\
&= \frac{1}{3} Var(X) + \frac{1}{2} \rho(X, Y) \sqrt{Var(X)} \sqrt{Var(Y)} \\
&= \frac{1}{3} \cdot 9 + \frac{1}{2} \cdot (-\frac{1}{2}) \cdot 3 \cdot 9 = 0.
\end{aligned}
$$

Thus, X and Z are independent. $\square$

The end

Thank you for your patience !